Realized Volatility Forecasting with Time Series Foundation Models

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Abstract

Realized volatility forecasting is central to risk management and portfolio allocation, yet no study has systematically assessed whether pretrained time series foundation models can match established econometric benchmarks for this task. We evaluate three zero-shot foundation models against six econometric specifications on 50 assets from the VOLARE dataset (equities, FX, futures; 2015–2026) at horizons of 1, 5, and 22 days, using Diebold–Mariano tests and Model Confidence Sets. The best foundation models match or exceed the best econometric benchmarks, but rankings depend on loss function, horizon, and asset class. At the daily horizon, Log-HAR retains a narrow QLIKE edge while foundation models dominate on MSE and out-of-sample R^2; at longer horizons, Moirai 2.0 leads across all metrics. Performance is heterogeneous across architectures: not all foundation models are competitive, and rankings shift across asset classes.

Keywords: Realized volatility, foundation models, zero-shot forecasting, HAR, model confidence set **JEL Codes:** C22, C53, C55, G17

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1 Introduction

Forecasting realized volatility is central to risk management, derivative pricing, and portfolio allocation. Since the work of Andersen and Bollerslev (1998) and Barndorff-Nielsen and Shephard (2002), realized volatility constructed from high-frequency returns has become the standard model-free measure of ex-post price variation. The Heterogeneous Autoregressive (HAR) model of Corsi (2009) is the dominant forecasting benchmark: its three-component structure, aggregating past realized volatility at daily, weekly, and monthly frequencies, provides a parsimonious approximation to the long-memory dynamics that characterize volatility. Extensions incorporating jump variation (Andersen et al., 2007), semivariance asymmetries (Patton and Sheppard, 2015), and measurement error corrections (Bollerslev et al., 2016) have yielded mixed practical gains (Clements and Preve, 2021; Branco et al., 2024). Machine learning methods, including neural networks (Bucci, 2020; Brini and Toscano, 2025), random forests (Luong and Dokuchaev, 2018), graph-based approaches (Zhang et al., 2024, 2025), and convolutional architectures (Moreno-Pino and Zohren, 2024), have also been applied, but a broader assessment by Christensen et al. (2023) concludes that machine learning offers incremental improvements rather than a paradigm shift.

A different class of models has emerged over the past two years: time series foundation models (TSFMs). These are large pretrained transformers (Vaswani et al., 2017) trained on billions of time series observations across domains, capable of producing forecasts for previously unseen series without any retraining, a capability known as zero-shot forecasting. Leading examples include Chronos (Ansari et al., 2024), Moirai (Woo et al., 2024), and Lag-Llama (Rasul et al., 2024), several of which have released second-generation versions with improved architectures (Ansari et al., 2025; Liu et al., 2025a). Surveys document the rapid pace of development in this area (Liang et al., 2024; Ye et al., 2024; Miller et al., 2024). Outside finance, Carriero et al. (2024) applied zero-shot language models to macroeconomic forecasting and found that they do not outperform traditional econometric baselines, raising the question of whether similar conclusions hold for financial time series.

Applications of TSFMs to financial time series remain limited. Goel et al. (2025) tested TimesFM on realized volatility for 20 stocks and found that fine-tuning was necessary for the model to compete with HAR; zero-shot performance was poor. Rahimikia et al. (2025) evaluated several TSFMs on daily stock returns and reported uniformly negative zero-shot results. However, realized volatility differs from returns in ways that may favor foundation models: it is strictly positive, exhibits well-documented mean reversion and long-memory dependence (Andersen et al., 2003), and follows a distribution closer to the macroeconomic and physical series on which TSFMs were pretrained. To our knowledge, no study has yet evaluated multiple foundation models on realized volatility with formal statistical testing.

This paper conducts the first systematic comparison of zero-shot TSFMs against established econometric benchmarks for realized volatility forecasting. We evaluate three TSFMs against six econometric specifications across 50 assets spanning three asset classes (equities, FX, futures) and three forecast horizons ($h = 1, 5, 22$ days), using the recently released VOLARE dataset (Cipollini et al., 2026). The zero-shot setting, with no domain-specific training, isolates the value of general time series pretraining. We apply Diebold–Mariano (DM) tests (Diebold and Mariano, 1995) and Model Confidence Set (MCS) analysis (Hansen et al., 2011) to control for multiple comparisons.

Zero-shot TSFMs are competitive with the best econometric benchmarks, contrasting with prior negative findings on daily returns and on TimesFM 1.0. Model rankings depend on the loss function and horizon: Log-HAR retains a narrow QLIKE advantage at $h = 1$ for equities, but TSFMs dominate on MSE, MAE, and out-of-sample R^2 at all horizons, and Moirai 2.0 achieves the lowest QLIKE at $h = 5$ and $h = 22$. MCS analysis shows that Moirai 2.0 enters the 90% MCS for every equity at $h = 5$ and for 97.5% at $h = 22$, while Log-HAR enters for 90% at $h = 1$. Rankings are not stable across asset classes, with different TSFMs dominating for equities, FX, and futures. Levels-based HAR extensions produce near-zero forecasts that inflate QLIKE, a practical limitation that log-space and TSFM approaches avoid.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the VOLARE dataset. Section 4 presents the econometric and foundation model specifications, along with the forecast evaluation framework. Section 5 reports the empirical results on univariate realized variance. Section 6 assesses statistical significance through DM tests, MCS, MZ regressions, and GR fluctuation tests. Section 7 extends the analysis to realized covariance forecasting and portfolio construction. Section 8 contains robustness checks, and Section 9 concludes.

2 Literature Review

2.1 Realized Volatility Modeling

The theory of realized volatility originates with Andersen and Bollerslev (1998) and Andersen et al. (2001), who showed that the sum of squared intraday returns provides a consistent, nonparametric estimator of the integrated variance of asset prices. Barndorff-Nielsen and Shephard (2002) formalized this result in a semimartingale framework, establishing that realized variance converges to integrated variance as the sampling frequency increases. Subsequent work identified the key stylized facts of realized volatility: approximate log-normality, long-memory dependence with a fractional integration parameter $d \approx 0.4$, and slow mean reversion (Andersen et al., 2003).

At ultra-high sampling frequencies, microstructure noise from bid-ask bounce and discrete price changes biases the realized variance estimator upward. Barndorff-Nielsen et al. (2008) introduced the realized kernel as a noise-consistent alternative that remains valid in the presence of market microstructure effects.

The HAR model of Corsi (2009) became the standard forecasting benchmark for realized volatility. By including daily, weekly, and monthly RV components as regressors, the model approximates long-memory behavior through a cascade of time scales corresponding to heterogeneous trading horizons. The specification is parsimonious (three regressors plus

a constant) yet achieves accuracy comparable to far more complex alternatives.

A large body of work has extended the HAR framework. The HAR-J model (Andersen et al., 2007) separates realized variance into continuous and jump components using bipower variation (Barndorff-Nielsen and Shephard, 2004). The HAR-RS model (Patton and Sheppard, 2015; Barndorff-Nielsen et al., 2010) decomposes realized variance into positive and negative semivariance to capture asymmetric responses to upside and downside moves. The HARQ model (Bollerslev et al., 2016) interacts the daily RV regressor with realized quarticity to account for time-varying measurement error. Clements and Preve (2021) provides a practical guide to implementing and comparing these extensions.

Long-memory models offer a complementary approach. ARFIMA models (Baillie et al., 1996; Izzeldin et al., 2019) explicitly parameterize the fractional integration order and remain competitive at longer forecast horizons where the slow mean reversion of realized volatility becomes the dominant dynamic.

Machine learning methods have been applied to realized volatility forecasting with mixed results. Bucci (2020) reports that feedforward and recurrent neural networks outperform HAR on a sample of U.S. stocks. Luong and Dokuchaev (2018) finds gains from random forests. Moreno-Pino and Zohren (2024) introduces DeepVol, a convolutional neural network that processes raw order book data. In contrast, Branco et al. (2024) concludes that simple linear models are difficult to beat after correcting for multiple testing, and Christensen et al. (2023) finds that machine learning offers only incremental improvements over standard econometric methods. Recent surveys by Gunnarsson et al. (2024) and Leushuis and Petkov (2026) cover this literature.

2.2 Time Series Foundation Models

Time series foundation models are large pretrained models that produce forecasts for arbitrary time series without task-specific training, analogous to large language models for text (Liang et al., 2024). TSFMs differ from task-specific deep learning models (neural networks,

recurrent networks, convolutional architectures) in a fundamental way: deep learning models must be trained from scratch on the target series before they can produce any forecast, while TSFMs are pretrained on large external corpora and can forecast immediately via zero-shot inference. This "plug-and-play" capability eliminates the need for domain-specific training data, hyperparameter tuning, and GPU-intensive estimation—a practical advantage when deploying to new assets or markets where historical data may be limited. TSFMs are trained on vast corpora of time series drawn from diverse domains (weather, energy, retail, transport, macroeconomics) and learn general temporal patterns that transfer to unseen series at inference time.

Three architectural families dominate the first generation of TSFMs. Chronos (Ansari et al., 2024) converts continuous values into discrete tokens via quantile binning and uses a T5 encoder-decoder architecture for autoregressive forecasting. T5, the Text-to-Text Transfer Transformer (Raffel et al., 2020), is a sequence-to-sequence model that reads the full input through an encoder and generates output tokens autoregressively through a decoder. Moirai (Woo et al., 2024) handles arbitrary numbers of variates through an "any-variate" attention mechanism and uses mixture distribution outputs for uncertainty quantification. Lag-Llama (Rasul et al., 2024) adapts the LLaMA decoder-only architecture for probabilistic time series forecasting using lag-based tokenization. Finance-specific foundation models have also been proposed, including Kronos (Shi et al., 2025), which is pretrained on candlestick (OHLCV) data from over 45 global exchanges using a learned tokenizer that maps price patterns into discrete tokens.[1]

Second-generation models appeared in 2025. Ansari et al. (2025) introduced Chronos-2 and Chronos-Bolt: the former adds multivariate support and covariate handling, while Chronos-Bolt is a distilled variant, meaning its parameters were trained to approximate the full Chronos-2 model's behavior at a fraction of the computational cost, producing direct

[1]We do not evaluate Kronos because it expects OHLCV price data as input and cannot be meaningfully applied to univariate realized variance without constructing synthetic candlestick data, which introduces an arbitrary input transformation unrelated to the model's intended use case.

quantile forecasts in a single forward pass and running up to 250 times faster than the original. Moirai 2.0 (Liu et al., 2025a) demonstrated that smaller, better-trained models can match or exceed their larger predecessors using multi-token prediction and improved tokenization. Other models include MOMENT (Goswami et al., 2024), a masked-encoder architecture designed for multiple time series tasks (forecasting, classification, anomaly detection), Timer (Liu et al., 2024), and Sundial (Liu et al., 2025b), each exploring different design choices for temporal representation and output distribution.

Standardized benchmarks have been developed to evaluate TSFMs across domains. GIFT-Eval (Aksu et al., 2024) provides a unified evaluation protocol across multiple datasets and forecast horizons. FEV-Bench (Shchur et al., 2025) focuses on forecasting evaluation with emphasis on calibration. TSFM-Bench (Li et al., 2024) compares models across zero-shot and fine-tuned settings. A consistent finding is that no single model dominates across all domains and horizons, which motivates domain-specific evaluations such as the one we undertake here. Adler et al. (2025) raises additional concerns about the calibration of TSFM prediction intervals, finding systematic miscoverage across several models.

2.3 Foundation Models for Volatility Forecasting

The application of TSFMs to volatility forecasting is recent and limited. The closest prior work is Goel et al. (2025), who tested TimesFM 1.0 on realized volatility for 20 stocks. They found that zero-shot TimesFM underperformed HAR and that fine-tuning was necessary to achieve competitive accuracy. Our paper differs in several respects: we evaluate three foundation models from different families (Chronos-Bolt, Moirai 2.0, and Lag-Llama), use newer model generations, and apply MCS analysis to control for multiple comparisons. The same research group also applied TimesFM to Value-at-Risk forecasting (Goel et al., 2024).

Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and found that zero-shot TSFMs consistently underperformed ARIMA. However, daily returns differ from realized volatility in fundamental ways: returns are approxi-

mately symmetric, near-white-noise, and lack the strong serial dependence that characterizes realized volatility. Realized volatility is strictly positive, mean-reverting, and displays long-memory dynamics, properties that bring it closer to the macroeconomic and physical time series on which TSFMs were pretrained. This distinction suggests that negative results on returns need not carry over to realized measures.

Other financial applications include stock price forecasting (Łaniewski and Ślepaczuk, 2025; Valeyre and Aboura, 2024), foreign exchange volatility modeling (Nguyen et al., 2025), and multivariate financial time series (Marconi, 2025). Finance-specific foundation models have also been proposed, including FinCast (Zhu et al., 2025) and Kronos (Shi et al., 2025).

The question of whether to fine-tune or apply TSFMs zero-shot is itself an active area of research. Das et al. (2025) explored in-context fine-tuning for TimesFM, while Germán-Morales et al. (2024) and Gupta et al. (2024) investigated parameter-efficient adaptation via Low-Rank Adaptation (LoRA) (Hu et al., 2022). Testing zero-shot performance first establishes an important baseline: if a TSFM already performs well without adaptation, the marginal value of fine-tuning may be limited, and the zero-shot workflow offers practical advantages in simplicity and computational cost.

We contribute to this emerging literature by providing the first evaluation that compares multiple time series foundation models against a broad set of econometric benchmarks for realized volatility, with formal statistical testing via DM tests and MCS.

3 The VOLARE Dataset

We use the VOLARE (VOLatility Archive for Realized Estimates) dataset of Cipollini et al. (2026), which provides daily realized volatility measures for a broad cross-section of financial assets. VOLARE is constructed from ultra-high-frequency tick data sourced from Kibot, covering 40 U.S. equities, 5 major currency pairs, and 5 commodity and index futures contracts. The equity sample begins on January 2, 2015 and runs through January 30, 2026,

yielding 2,786 trading days per stock. The FX and futures samples begin on September 25, 2009 (4,242 and 4,228 trading days, respectively).

For each asset-day, VOLARE provides realized measures computed at multiple sampling frequencies. All measures used in this paper are computed at the 5-minute sampling frequency: realized variance (RV), bipower variation (BPV), positive and negative realized semivariance (RS^+, RS^-), and realized quarticity (RQ). All values are expressed in decimal squared returns; for instance, a typical daily RV for a U.S. equity is approximately 1.9×10^{-4}, corresponding to an annualized volatility of roughly 22% ($\sqrt{252 \times 1.9 \times 10^{-4}} \approx 0.22$). VOLARE is well suited for our multi-asset comparison because it constructs all realized measures using a uniform methodology across assets, ensuring that cross-asset differences in model rankings reflect genuine forecasting performance rather than measurement inconsistencies. In addition to univariate realized measures, VOLARE provides daily realized covariance matrices for each asset class, computed from 5-minute intraday cross-products of returns. These matrices are symmetric and positive semi-definite by construction, and we use them in Section 7 for covariance forecasting and portfolio applications.

Our equity sample comprises the 40 stocks in VOLARE that span the full 2015–2026 period: AAPL, ADBE, AMD, AMGN, AMZN, AXP, BA, CAT, CRM, CSCO, CVX, DIS, GE, GOOGL, GS, HD, HON, IBM, JNJ, JPM, KO, MCD, META, MMM, MRK, MSFT, NFLX, NKE, NVDA, ORCL, PG, PM, SHW, TRV, TSLA, UNH, V, VZ, WMT, and XOM. These cover all 11 Global Industry Classification Standard (GICS) sectors. The FX sample consists of five major currency pairs (AUDUSD, EURUSD, GBPUSD, USDCAD, USDJPY) and the futures sample covers five contracts (Corn, Crude Oil, E-mini S&P 500, Gold, Natural Gas).

Table 1 reports descriptive statistics for the 5-minute realized variance across the three asset classes. As expected for a strictly positive quantity, realized variance is right-skewed across all asset classes, with heavy tails (kurtosis ranging from 35 to over 3,000) confirming that extreme volatility episodes are a persistent feature of the data. The first-order

autocorrelation of daily RV, denoted ρ_1, averages 0.606 for equities, consistent with the well-documented persistence of volatility; the 22-day autocorrelation (ρ_{22}, i.e., the sample autocorrelation at lag 22) averages 0.085, indicating slow decay. The FX pairs display RV approximately two orders of magnitude smaller than equities (cross-sectional average $\bar{RV} \approx 3 \times 10^{-5}$ vs. 2×10^{-4}), with comparable persistence (cross-sectional average $\bar{\rho}_1 = 0.52$, where the bar denotes averaging across assets). Futures exhibit the widest cross-asset heterogeneity: the E-mini S&P 500 (ES) has $\rho_1 = 0.779$, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$), presenting a natural stress test for forecasting models.

	N	Mean	Median	Skew	Kurt	ρ_1	ρ_{22}
Panel A: Equities (cross-sectional average, 40 stocks)							
Average	2,786	1.91	1.12	13.5	269	0.606	0.085
Panel B: Foreign Exchange (5 pairs)							
AUDUSD	4,241	0.48	0.37	11.3	232	0.577	0.138
EURUSD	4,241	0.28	0.22	5.0	51	0.570	0.263
GBPUSD	4,242	0.32	0.24	27.5	1,117	0.484	0.061
USDCAD	4,240	0.25	0.19	4.3	35	0.654	0.368
USDJPY	4,239	0.33	0.22	14.4	349	0.325	0.063
Panel C: Futures (5 contracts)							
Corn (C)	4,185	2.48	1.54	12.3	238	0.178	0.079
Crude Oil (CL)	4,223	6.87	3.87	57.6	3,546	0.195	0.047
E-mini S&P (ES)	4,224	1.11	0.53	12.9	212	0.779	0.110
Gold (GC)	4,206	3.63	2.32	62.3	3,963	−0.001	−0.000
Natural Gas (NG)	4,221	10.30	6.35	19.0	522	0.187	0.267

Table 1: Descriptive statistics for daily realized variance measured at the 5-minute frequency. The equity panel reports cross-sectional averages across 40 stocks; FX and futures report individual assets. ρ_k denotes the sample autocorrelation at lag k (e.g., ρ_{22} is the autocorrelation at 22 trading days). All RV values are in decimal squared returns ($\times 10^{-4}$).

Table 2 reports descriptive statistics for the realized covariance matrices across the three asset classes. Mean diagonal elements (realized variances) are consistent with the univariate statistics in Table 1. Off-diagonal elements are predominantly positive for stocks (mean covariance 0.66×10^{-4}), reflecting equity co-movement, and near zero for FX (-0.008×10^{-4}), reflecting offsetting currency movements. The median condition number of 131.5 for the

40-stock matrices is substantially higher than for the 5-asset FX (11.6) and futures (20.7) matrices, foreshadowing the dimensionality challenges for portfolio construction documented in Section 7. All matrices are positive semi-definite by construction.

	Stocks	FX	Futures
Panel A: Sample characteristics			
Assets (N)	40	5	5
Trading days (T)	2,786	4,242	4,223
Unique pairs $N(N+1)/2$	820	15	15
Sample period	2015–2026	2009–2026	2009–2026
Panel B: Diagonal elements (variances, $\times 10^{-4}$)			
Mean	2.732	0.342	4.974
Median	1.411	0.241	1.765
Std. dev.	6.607	0.511	73.220
Panel C: Off-diagonal elements (covariances, $\times 10^{-4}$)			
Mean	0.660	−0.008	0.230
Median	0.254	−0.025	0.068
Std. dev.	1.950	0.241	0.957
Panel D: Matrix properties			
Median condition number	131.5	11.6	20.7
Fraction PSD (%)	100.0	100.0	100.0

Table 2: Descriptive statistics for daily realized covariance matrices

Notes: Descriptive statistics for daily realized covariance matrices constructed from ultra-high-frequency data (VOLARE). Diagonal elements are realized variances; off-diagonal elements are realized covariances between asset pairs. All variance and covariance statistics are reported in units of 10^{-4}. The condition number is the ratio of the largest to smallest eigenvalue; we report the median across all trading days (the mean is dominated by occasional near-singular matrices). PSD denotes the fraction of matrices with all eigenvalues $\geq -10^{-12}$.

4 Methodology

4.1 Econometric Benchmarks

We consider six econometric specifications that span the main approaches to realized volatility forecasting: the HAR family and its extensions, which exploit multi-scale aggregation, and ARFIMA, which explicitly models long memory.

HAR Model. The HAR model captures the multi-scale persistence of realized volatility by aggregating past RV at daily, weekly, and monthly frequencies:

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \tag{1}$$

where $RV_{t-k:t} \equiv k^{-1} \sum_{i=0}^{k-1} RV_{t-i}$ denotes the average realized volatility over the previous k days. We use the original specification, where the weekly and monthly components include overlapping lags (i.e., the weekly component averages days t through $t-4$, not the non-overlapping "rotated" version that separates lags 2–5 from 6–22). This three-component structure approximates long-memory behavior through the actions of heterogeneous market participants operating at different time horizons.

HAR-J Model. The HAR-J model augments the baseline HAR with a jump component to separate continuous and discontinuous variation:

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \beta_J J_t + \varepsilon_{t+h}, \tag{2}$$

where $J_t = \max(RV_t - BPV_t, 0)$ measures the jump component as the positive part of the difference between realized variance and bipower variation (Barndorff-Nielsen and Shephard, 2004). A negative coefficient on J_t indicates that large jumps reduce, rather than increase, future volatility.

HAR-RS Model. The HAR-RS model decomposes realized variance into positive and negative semivariance components to capture asymmetric volatility responses:

$$RV_{t+h} = \beta_0 + \beta_1^+ RS_t^+ + \beta_1^- RS_t^- + \beta_2^+ RS_{t-5:t}^+ + \beta_2^- RS_{t-5:t}^- + \beta_3^+ RS_{t-22:t}^+ + \beta_3^- RS_{t-22:t}^- + \varepsilon_{t+h}, \tag{3}$$

where RS_t^+ and RS_t^- denote good and bad realized semivariance, respectively, and $RS_t^+ + RS_t^- = RV_t$ by construction (Barndorff-Nielsen et al., 2010). The six-regressor structure

allows upside and downside risk to follow separate dynamics at each aggregation frequency.

HARQ Model. The HARQ model accounts for time-varying measurement error in realized volatility by interacting the daily RV regressor with the square root of realized quarticity:

$$RV_{t+h} = \beta_0 + (\beta_1 + \beta_1^Q \sqrt{RQ_t})\, RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \tag{4}$$

where RQ_t is the realized quarticity. The interaction term attenuates the daily RV signal when measurement noise is high, as indicated by large values of RQ_t.

Log-HAR Model. The Log-HAR model applies the HAR specification to log-transformed realized volatility:

$$\log RV_{t+h} = \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_{t-5:t} + \beta_3 \log RV_{t-22:t} + \varepsilon_{t+h}. \tag{5}$$

The logarithmic transformation brings the RV distribution closer to normality, improving OLS efficiency. Point forecasts in levels are recovered via bias-corrected retransformation: $\widehat{RV}_{t+h} = \exp(\hat{y}_{t+h} + \hat{\sigma}^2/2)$, where $\hat{\sigma}^2$ is the estimated residual variance, following the standard log-normal adjustment.

ARFIMA Model. The ARFIMA model captures the well-documented long-memory property of realized volatility through fractional differencing:

$$(1-L)^d\, \Phi(L)\, (\log RV_t - \mu) = \Theta(L)\, \varepsilon_t, \tag{6}$$

where $d \in (0, 0.5)$ is the fractional integration parameter, $\Phi(L)$ and $\Theta(L)$ are autoregressive and moving average lag polynomials of orders p and q (with $p, q \leq 2$ selected by AIC), and L is the lag operator. The parameter d is estimated by maximum likelihood. The fractional differencing filter is truncated at $K = 100$ lags.

All HAR-family models are estimated by OLS with Newey–West heteroskedasticity and autocorrelation consistent standard errors. The ARFIMA model is re-estimated every 22 trading days to balance computational cost against parameter staleness.

4.2 Time Series Foundation Models

We evaluate three time series foundation models (TSFMs) in a zero-shot setting, applied directly to realized volatility series without any domain-specific training or fine-tuning. The zero-shot approach is the most practically relevant deployment mode: it requires no labeled financial data, no model retraining, and can be applied immediately to any new asset. Among the three TSFMs we evaluate, Chronos-Bolt and Moirai 2.0 are inference-only architectures with no public fine-tuning API. Lag-Llama supports fine-tuning through its GluonTS estimator interface, and the full Chronos-2 model (distinct from the distilled Bolt variant) provides built-in LoRA fine-tuning via its `fit()` method; we leave these adaptation strategies for future work.[2][3][4]

Formal framework. A TSFM can be expressed as a parametric mapping from an observed history to a forecast distribution. Let $y_{1:T} = (y_1, \ldots, y_T)$ denote the context window of T past observations (in our case, $T = 512$ daily RV values). A TSFM with pretrained parameters $\hat{\theta}$ produces a forecast of the next H values:

$$\hat{y}_{T+1:T+H} = f_{\hat{\theta}}(y_{1:T}), \tag{7}$$

[2]We originally planned to include TimesFM 2.5 (Das et al., 2024), but its publicly released model weights are incompatible with the current version of the `timesfm` Python package (v1.3.0), producing dimension mismatches during inference that prevent reliable forecasting.

[3]We do not evaluate MOMENT (Goswami et al., 2024) because it is pretrained with a masked reconstruction objective rather than an autoregressive forecasting objective. Its forecasting head, a linear projection layer that maps patch embeddings to the forecast horizon, is not pretrained and must be trained on the target series before the model can produce any predictions (Goswami et al., 2024, Section 4.2). This makes it incompatible with our zero-shot evaluation protocol.

[4]Carriero et al. (2024) additionally evaluate Tiny Time Mixers (TTM) (Ekambaram et al., 2024) and TimeGPT (Garza et al., 2024) for macroeconomic forecasting. We exclude TTM due to software version conflicts with our inference environment and TimeGPT because it is a proprietary, closed-source API that does not permit inspection of model weights or training data, precluding reproducibility.

where $f_{\hat{\theta}}$ maps the input sequence to a predictive distribution over future values.[5] The parameters $\hat{\theta}$ are learned during a pretraining phase on large external corpora of time series spanning weather, energy, retail, transport, and macroeconomic domains. At inference time, the model receives only the context window $y_{1:T}$ and produces forecasts without any task-specific parameter updates—this is the zero-shot setting.

TSFM components. Every TSFM implements the mapping in Equation (7) through three stages: *tokenization*, *encoding*, and *decoding*. First, the raw series $y_{1:T}$ is converted into a sequence of discrete or continuous *tokens*—the model's internal representation of the data. Tokenization serves the same role as variable construction in an econometric model: it determines which features of the history the model can "see." The three TSFMs we evaluate use different tokenization strategies (quantile binning, patch projection, lag features), summarized in Table 3 and described individually below. Second, the token sequence is processed by a *transformer* (Vaswani et al., 2017), a neural network in which each position attends to other positions via learned attention weights, capturing dependencies at arbitrary lags without imposing the specific functional form that HAR prescribes (daily, weekly, monthly components). Third, the transformer's output is passed through a *decoding head* that maps internal representations back to forecasts in the original scale. Decoding heads differ across models: some output quantiles of the predictive distribution directly, others parameterize an explicit probability distribution (e.g., Student-t) from which forecast trajectories are sampled. In both cases, we extract the median as the point forecast, analogous to reporting the conditional median from a quantile regression.

[5] More precisely, $f_{\hat{\theta}}$ outputs a distribution $\hat{p}(y_{T+1:T+H} \mid y_{1:T}; \hat{\theta})$ from which we extract the median as the point forecast. The parameters $\hat{\theta}$ are fixed at their pretrained values throughout our evaluation; no gradient updates occur during inference.

Model	Architecture	Tokenization	Params	Output	Release
Chronos-Bolt-S	Encoder-only	Quantile binning	48M	Direct quantiles	2025
Chronos-Bolt-B	Encoder-only	Quantile binning	205M	Direct quantiles	2025
Moirai 2.0-S	Decoder-only	Patch projection	14M	9 quantile levels	2025
Lag-Llama	Decoder-only	Lag features	7.5M	Student-t mixture	2024

Table 3: Summary of TSFM architectures evaluated. All models are applied zero-shot with a 512-day context window. "Params" refers to the total number of pretrained parameters. The point forecast is the median of each model's predictive distribution.

Chronos-Bolt (Ansari et al., 2025). Chronos-Bolt is a distilled variant of the Chronos family. The original Chronos model tokenizes continuous values by mapping them into 4,096 discrete bins via quantile binning, where each observation is replaced by the index of its nearest quantile in a reference distribution, much as one would discretize a continuous variable into histogram bins. The model then processes these token indices through a T5 encoder-decoder transformer and generates forecasts autoregressively, predicting one token at a time. Chronos-Bolt replaces this two-step architecture with an encoder-only transformer that reads the entire input context and produces all forecast quantiles in a single forward pass, without iterative decoding. This yields a 250× speedup over the original while maintaining competitive accuracy. The decoding head outputs quantile forecasts directly (e.g., the 10th, 50th, and 90th percentiles), bypassing the need to sample from a token distribution. We use both the small (48M parameters) and base (205M parameters) checkpoints[6] and take the median (0.5 quantile) as the point forecast.

Moirai 2.0 (Liu et al., 2025a). Moirai 2.0 builds on the original Moirai architecture. It tokenizes the input series by dividing it into fixed-length *patches* (contiguous blocks of consecutive observations, e.g., 32 days each) and projects each patch into the transformer's embedding space via a learned linear map. This is analogous to constructing non-overlapping moving-window summaries of the series, except that the summary is learned rather than

[6] `amazon/chronos-bolt-small` and `amazon/chronos-bolt-base` on HuggingFace.

hand-specified. The transformer is decoder-only: it processes patches causally (each patch attends only to itself and earlier patches), preserving the temporal ordering. Moirai 2.0 uses *multi-token prediction*, generating multiple future patches in a single pass rather than one step at a time. The decoding head outputs nine quantile levels (0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95) per forecast step. We use the small checkpoint[7] and extract the median quantile as the point forecast.

Lag-Llama (Rasul et al., 2024). Lag-Llama is a decoder-only foundation model based on the LLaMA architecture, adapted for univariate probabilistic time series forecasting. Unlike the patch-based models above, Lag-Llama tokenizes each time step individually using a set of *lag features*: at each position t, the input vector consists of the current value and a fixed set of lagged values $(y_{t-1}, y_{t-2}, y_{t-7}, y_{t-14}, \ldots)$, concatenated with time-related covariates. This lag-based design is conceptually closer to an autoregressive econometric specification, since the model explicitly receives lagged values as inputs, though the transformer learns the mapping from lags to forecasts nonparametrically rather than through a linear projection. The decoding head parameterizes a Student-t mixture distribution at each forecast step: it outputs the location, scale, and degrees-of-freedom parameters for each mixture component. Forecasts are generated by sampling trajectories from this distribution and taking the median across samples. We use the pretrained checkpoint (8 layers, 4 attention heads, 36-dimensional embeddings per head), sample 20 trajectories, and report the median as the point forecast.

Zero-shot protocol. All three TSFMs receive a rolling context window of 512 daily RV observations as input, with no additional features or covariates. The 512-day context length is the maximum supported by all three architectures' positional encodings and is the default recommended setting; it corresponds to approximately two years of trading data, providing enough history to capture the long-memory dynamics of realized volatility. By contrast, the 252-day rolling estimation window for econometric models follows the convention of using one

[7] `Salesforce/moirai-2.0-R-small` on HuggingFace.

year of daily data, which is standard in the HAR literature. The one-year window balances the need for sufficient data to estimate HAR's four parameters reliably against the risk of including stale observations from a different volatility regime. This reflects each approach's recommended configuration rather than an equal-information comparison; we view this as the practically relevant setup, since a practitioner would deploy each model with its standard settings. We assess sensitivity to the context window length in Section 8.

For multi-step horizons ($h > 1$), the point forecast is the mean of the h-step output vector. No parameters are updated during the out-of-sample evaluation period; all forecasts are purely zero-shot. We extract the median rather than the mean of each model's predictive distribution because the median is the optimal point forecast under MAE loss and is more robust to the right-skewed forecast distributions that arise in volatility applications; for symmetric distributions, the two coincide. We acknowledge that the median is not the optimal point forecast under QLIKE loss (the conditional mean is); however, in practice, the median is the standard output of probabilistic TSFMs and produces more stable forecasts than the sample mean when the predictive distribution is heavy-tailed. Table 4 reports wall-clock timing for all models. Econometric specifications require < 0.01 s per forecast step, run entirely on CPU, and impose no special hardware requirements; they execute on commodity hardware including a standard laptop. TSFM inference was conducted on the Duke Computing Cluster (DCC) using a single NVIDIA RTX A5000 GPU (24 GB VRAM), with per-step times ranging from 0.014 s (Chronos-Bolt-Small) to 0.084 s (Lag-Llama).

4.3 Pretraining Data and Contamination Risk

A natural concern with zero-shot evaluation is whether TSFM performance is inflated because the models' pretraining corpora overlap with the test series. Table 5 summarizes each model's training data and its financial content. No TSFM in our study was trained on realized volatility. VOLARE's realized variance series are second-moment statistics derived from ultra-high-frequency intraday returns at 5-minute sampling (Cipollini et al., 2026); this

Model	Parameters	Device	Sec/step (h=1)	Total (1 asset)
Econometric models				
HAR	4	CPU	0.003	21 s
HAR-J	5	CPU	0.003	21 s
HAR-RS	7	CPU	0.004	27 s
HARQ	5	CPU	0.004	24 s
Log-HAR	4	CPU	0.004	30 s
ARFIMA	$\leq$7	CPU	0.006	45 s
Time series foundation models				
Chronos-Bolt-S	48M	GPU	0.014	1.6 min
Chronos-Bolt-B	205M	GPU	0.025	2.8 min
Moirai-2.0-S	14M	GPU	0.019	2.2 min
Lag-Llama	8M	GPU	0.084	9.5 min

Table 4: Computational cost per forecast step

Computational cost per forecast step. TSFM inference was run on the Duke Computing Cluster (DCC) using SLURM job scheduling on the `gpu-common` partition, with a single NVIDIA RTX A5000 GPU (24 GB). Econometric models require only CPU and run on commodity hardware (including a standard laptop) in under one minute per asset; their timings here were measured on a DCC node (Intel Xeon Gold 5320, 2.20 GHz). Seconds per step is the wall-clock time to produce one forecast for one asset-day, averaged across the equity sample ($n \approx$ 2,274 out-of-sample steps). Total time is the approximate wall-clock for one asset across three horizons. ARFIMA is re-estimated every 22 steps. Lag-Llama timing includes 20 sample paths per step.

quantity does not appear in any known training corpus.

Table 5: TSFM pretraining corpora and financial data exposure. "Financial Share" reports the fraction of total training observations from financial or economic domains. No model was trained on realized volatility or any intraday-derived statistic.

Model	Corpus Size	Financial Share	Financial Datasets	RV in Corpus?
Chronos-Bolt	∼890K series, 84B obs	<0.3%	`exchange_rate`, FRED-MD	No
Moirai 2.0	27.6B obs (LOTSA)	0.10%	Undisclosed	No
Lag-Llama	∼8K series	Small	`exchange_rate`	No

Financial data constitutes less than 1% of training observations for all three TSFMs we evaluate—Chronos-Bolt (Ansari et al., 2024), Moirai 2.0 (Liu et al., 2025a), and Lag-Llama (Rasul et al., 2024)—limited to daily exchange rates and macroeconomic indicators. This minimal financial exposure makes data contamination an unlikely explanation for TSFM performance on realized volatility. Instead, the results suggest that TSFM accuracy reflects transfer of general temporal patterns (mean reversion, long memory, regime persistence) rather than memorization of financial dynamics.

4.4 Forecast Evaluation

Walk-forward design. We employ a walk-forward evaluation scheme with a rolling origin. Econometric models use a fixed estimation window of 252 trading days (approximately one year): at each origin, the model is estimated on the most recent 252 observations, and the window slides forward one day at a time, producing daily re-estimated forecasts. Foundation models follow the same walk-forward timing but use a 512-day context window and require no estimation step; only the context window slides forward. Because the TSFM context window is longer, the first TSFM forecast is produced 260 days later than the first econometric forecast; all model comparisons are conducted on the common out-of-sample period where both econometric and TSFM forecasts are available. For the equity sample (2,786 days), this produces approximately 2,148 common out-of-sample forecasts per model–asset combination at $h = 1$; the FX and futures samples (4,240+ days) yield approximately 3,602.

Forecast horizons. We evaluate three forecast horizons: $h = 1$ (one day), $h = 5$ (one week), and $h = 22$ (one month). The forecast target at horizon h is the average realized volatility over the next h days, $h^{-1} \sum_{i=1}^{h} RV_{t+i}$. All models produce direct h-step forecasts. For HAR-family models, this means estimating a separate regression for each horizon, where the dependent variable is the h-day average of future RV; the model does not iterate one-step-ahead predictions forward. ARFIMA similarly produces direct multi-step forecasts from its estimated ARMA representation on log RV. For TSFMs, each model generates a full forecast trajectory of length h in a single inference call; the point forecast is the mean of this h-step output vector. This direct approach avoids the compounding of errors that arises in iterated multi-step forecasting.

Loss functions and statistical testing. Our primary loss function is the quasi-likelihood (QLIKE) loss of Patton (2011):

$$L_{\text{QLIKE}}(\hat{\sigma}_t^2, RV_t) = \frac{RV_t}{\hat{\sigma}_t^2} - \log \frac{RV_t}{\hat{\sigma}_t^2} - 1, \tag{8}$$

where $\hat{\sigma}_t^2$ is the forecast and RV_t is the realized value. QLIKE preserves the ranking of forecasts regardless of the choice of volatility proxy, making it the preferred metric when the true conditional variance is unobserved. We also report mean squared error (MSE), mean absolute error (MAE), and the out-of-sample R^2 defined as $R^2_{\text{OOS}} = 1 - \sum_t (RV_t - \hat{f}_t)^2 / \sum_t (RV_t - \bar{RV})^2$, where $\bar{RV}$ is the historical mean of realized variance over the estimation window. Unlike the in-sample R^2, R^2_{OOS} can take negative values, indicating that the model's forecasts are less accurate than a simple historical mean forecast. All forecasts are floored at 10^{-6} to prevent numerical instability in the QLIKE calculation when models produce near-zero predictions.

Pairwise forecast comparisons use the DM test with Newey–West variance estimation, applied to QLIKE loss differentials. To address the multiple comparison problem inherent in evaluating eleven models, we compute the MCS, which identifies the subset of models

whose forecasting ability cannot be statistically distinguished from the best model at a given significance level. We use the $T_{\max}$ test statistic with a block bootstrap (block length = 22 days, $B = 10{,}000$ replications) and significance level $\alpha = 0.10$.

We assess forecast efficiency using the Mincer–Zarnowitz (MZ) regression (Mincer and Zarnowitz, 1969):

$$RV_t = \alpha + \beta \hat{f}_t + \varepsilon_t, \tag{9}$$

where $\hat{f}_t$ is the model's forecast. Under forecast optimality, $\alpha = 0$ and $\beta = 1$, meaning the forecast is unbiased and captures the correct scale of variation. We test the joint null $H_0 \colon \alpha = 0, \beta = 1$ using a Wald test with Newey–West standard errors and report cross-asset averages and rejection rates at the 5% level.

To examine whether relative forecast performance is stable over time, we apply the Giacomini and Rossi (2010) fluctuation test (GR). For each model paired against the HAR benchmark, we compute a rolling DM statistic over a window of size $m = \lfloor 0.3 \times T \rfloor$, producing a time path of relative QLIKE performance. The test statistic is $\sup_t |S_t|$, compared against critical values from the distribution of the supremum of a standardized Brownian bridge. Rejection of the null indicates that the two models' relative performance changes significantly over time.

5 Empirical Results

This section presents the univariate realized variance forecasting results. We first examine aggregate forecast accuracy across loss functions and horizons, then analyze cross-asset and cross-market heterogeneity.

5.1 Aggregate Forecast Accuracy

This subsection reports the overall forecasting performance averaged across assets within each asset class, without examining individual asset-level variation. The goal is to estab-

lish the broad ranking of models by loss function and horizon before turning to cross-asset heterogeneity and statistical testing in subsequent subsections.

Table 6 reports the cross-sectional average of per-asset loss functions across 40 equities. Figure 1 illustrates the forecasts of three representative models against realized values for four assets spanning equities and FX. The results reveal a systematic divergence between QLIKE and the other metrics.

Table 6: Forecast accuracy for 40 U.S. equities (VOLARE). Values are cross-sectional averages of per-asset loss functions. Bold indicates the best value in each column within each panel. † denotes inflated QLIKE due to near-zero forecasts or systematic forecast bias.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.244	1.43	1.38†	0.335
HAR-J	0.242	1.42	1.16†	0.349
HAR-RS	0.265	1.52	3.73†	0.268
HARQ	0.297	1.49	16.46†	0.111
Log-HAR	0.257	1.31	**0.220**	0.238
ARFIMA	0.295	1.63	0.314	0.153
Chronos-Bolt-S	0.216	1.18	0.229	0.408
Chronos-Bolt-B	0.213	**1.17**	0.230	0.420
Moirai-2.0-S	**0.210**	1.19	0.226	**0.426**
Lag-Llama	0.291	1.65	0.402	0.127
Panel: $h = 5$ (5 days)				
HAR	0.163	1.40	2.70†	0.209
HAR-J	0.159	1.39	2.47†	0.231
HAR-RS	0.184	1.50	7.04†	0.101
HARQ	0.192	1.43	16.16†	0.024
Log-HAR	0.164	1.28	0.231	0.135
ARFIMA	0.170	1.44	0.244	0.178
Chronos-Bolt-S	0.127	1.09	0.213	0.355
Chronos-Bolt-B	0.128	1.09	0.216	0.346
Moirai-2.0-S	**0.122**	**1.07**	**0.188**	**0.384**
Lag-Llama	0.181	1.43	0.305	0.044
Panel: $h = 22$ (22 days)				
HAR	0.117	1.47	6.06†	-0.031
HAR-J	0.117	1.46	5.92†	-0.015
HAR-RS	0.130	1.57	9.40†	-0.145
HARQ	0.131	1.48	12.42†	-0.184
Log-HAR	0.115	1.35	0.350	-0.044
ARFIMA	0.119	1.35	0.275	-0.014
Chronos-Bolt-S	0.100	1.13	0.322	0.085
Chronos-Bolt-B	0.101	1.13	0.325	0.077
Moirai-2.0-S	**0.098**	**1.07**	**0.257**	**0.110**
Lag-Llama	0.109	1.25	0.283	0.009

† Marks QLIKE > 1. For HAR-RS and HARQ, this reflects near-zero forecasts from levels-based OLS.

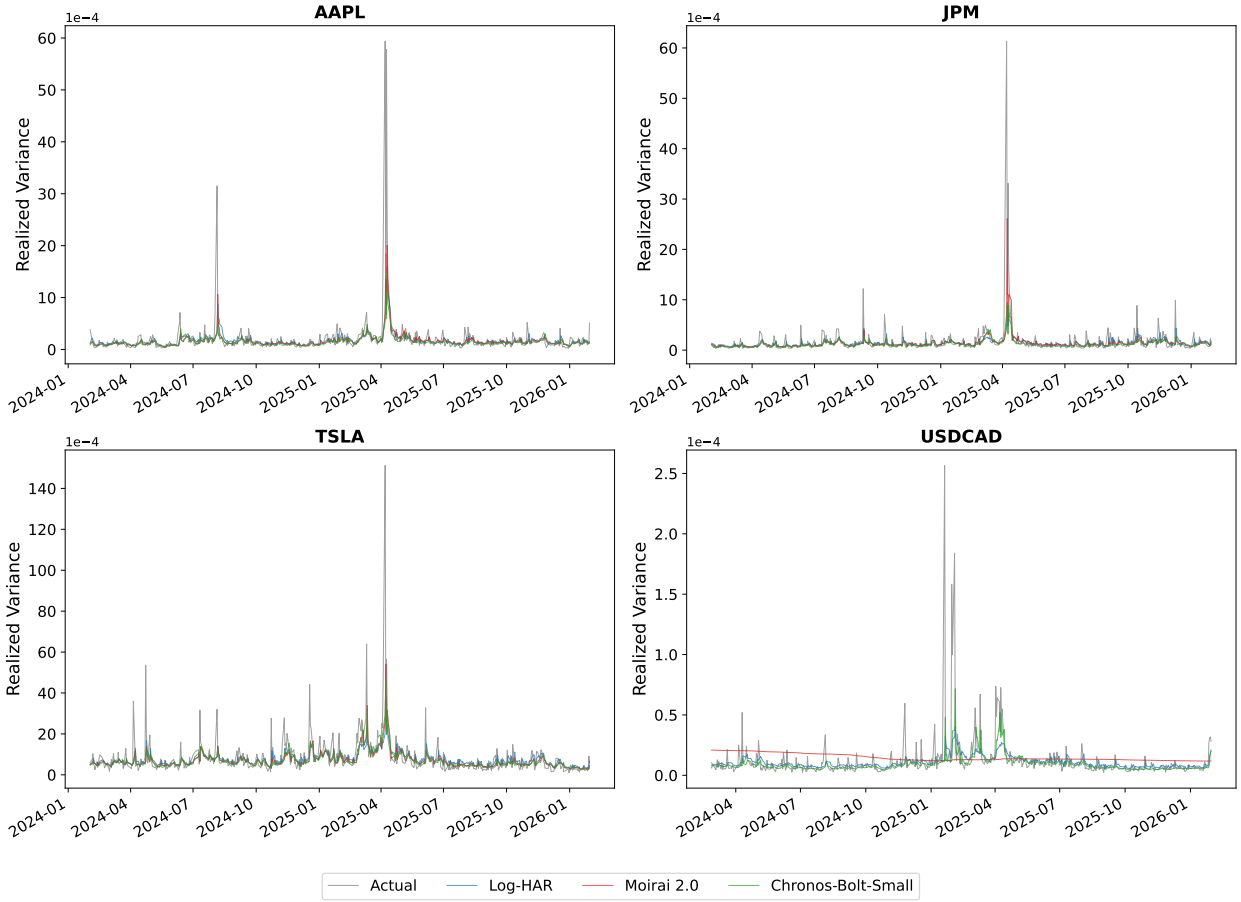

Figure 1: Forecast vs. actual realized variance at $h = 1$ for four representative assets. The last 500 out-of-sample observations are shown. Gray: actual RV; blue: Log-HAR; red: Moirai 2.0; green: Chronos-Bolt-Small.

At the one-day horizon ($h = 1$), Log-HAR achieves the lowest average QLIKE (0.220), followed closely by Moirai 2.0 (0.226) and Chronos-Bolt-Small (0.229). The gap between Log-HAR and the best TSFM is small (roughly 3% in relative terms) and is driven by the log-space formulation of Log-HAR, which guarantees positive forecasts and penalizes proportional forecast errors in a way that aligns with QLIKE's structure. On MSE, MAE, and R^2_{OOS}, however, the TSFMs dominate: Moirai 2.0 achieves the lowest MSE (0.210×10^{-6}), and the highest R^2_{OOS} (0.426), compared to 0.238 for Log-HAR. Chronos-Bolt-Base achieves the lowest MAE (1.17×10^{-4}). The out-of-sample R^2 gap between TSFMs and the best

econometric model is 8–19 percentage points at $h = 1$, a substantial margin.

At the five-day horizon ($h = 5$), the TSFM advantage becomes more pronounced. Moirai 2.0 achieves the best score across all four metrics: MSE (0.122×10^{-6}), MAE (1.07×10^{-4}), QLIKE (0.188), and R^2_{OOS} (0.384). The gap over Log-HAR in QLIKE terms is now 19%, compared to 3% at $h = 1$. Chronos-Bolt-Small achieves the second-lowest QLIKE (0.213), still below Log-HAR's 0.231. The econometric models that do best at $h = 1$ (HAR, HAR-J) deteriorate more rapidly at longer horizons, while the TSFMs maintain relatively stable accuracy.

At the monthly horizon ($h = 22$), Moirai 2.0 again dominates: QLIKE = 0.257, MSE = 0.098×10^{-6}, and $R^2_{\text{OOS}} = 0.110$. The econometric benchmarks now show near-zero or negative R^2_{OOS}, indicating that they barely outperform (or underperform) the historical mean at this horizon. Log-HAR achieves $R^2_{\text{OOS}} = -0.044$, while Chronos-Bolt-Small maintains a positive R^2_{OOS} of 0.085. This horizon-dependent pattern is consistent with the hypothesis that TSFMs' large context windows (512 days) capture long-range dependencies more effectively than the HAR's fixed three-component structure.

Table 7: Forecast accuracy for 40 U.S. equities (VOLARE), cross-sectional medians. Median aggregation is robust to outlier assets with degenerate forecasts. Bold indicates the best value in each column within each panel.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.116	1.14	0.221	0.352
HAR-J	0.113	1.14	0.567	0.365
HAR-RS	0.124	1.20	$1.37^{\dagger}$	0.318
HARQ	0.173	1.24	$11.12^{\dagger}$	0.066
Log-HAR	0.139	1.12	**0.208**	0.237
ARFIMA	0.147	1.37	0.297	0.163
Chronos-Bolt-S	0.096	0.97	0.212	0.436
Chronos-Bolt-B	0.094	**0.95**	0.213	**0.459**
Moirai-2.0-S	**0.091**	0.98	0.208	0.451
Lag-Llama	0.156	1.38	0.384	0.134
Panel: $h = 5$ (5 days)				
HAR	0.096	1.13	0.269	0.253
HAR-J	0.094	1.12	0.384	0.271
HAR-RS	0.109	1.19	$3.59^{\dagger}$	0.179
HARQ	0.126	1.17	$13.39^{\dagger}$	0.017
Log-HAR	0.111	1.08	0.205	0.133
ARFIMA	0.098	1.19	0.232	0.192
Chronos-Bolt-S	0.077	0.90	0.196	0.366
Chronos-Bolt-B	0.080	0.91	0.196	0.359
Moirai-2.0-S	**0.073**	**0.86**	**0.172**	**0.383**
Lag-Llama	0.128	1.20	0.295	0.039
Panel: $h = 22$ (22 days)				
HAR	0.078	1.23	$3.40^{\dagger}$	0.007
HAR-J	0.076	1.22	$3.39^{\dagger}$	0.016
HAR-RS	0.082	1.30	$5.23^{\dagger}$	-0.090
HARQ	0.089	1.25	$9.29^{\dagger}$	-0.121
Log-HAR	0.081	1.13	0.296	-0.043
ARFIMA	0.075	1.10	0.269	0.073
Chronos-Bolt-S	0.071	0.94	0.319	0.085
Chronos-Bolt-B	0.072	0.97	0.326	0.082
Moirai-2.0-S	**0.069**	**0.89**	**0.248**	**0.112**
Lag-Llama	0.079	1.03	0.267	0.002

Table 7 reports cross-sectional medians, which eliminate the influence of outlier assets where levels-based HAR variants produce degenerate forecasts. Under median aggregation, the qualitative model ranking is preserved: Log-HAR and Moirai 2.0 are tied for the best

QLIKE at $h = 1$ (both 0.208), Moirai 2.0 dominates at $h = 5$ (0.172) and $h = 22$ (0.248). The gap between TSFMs and econometric models narrows under medians — the median QLIKE difference between Log-HAR and Moirai 2.0 at $h = 5$ shrinks to 0.033 (vs. 0.043 in means) — confirming that the main findings are not driven by a few outlier assets. The median results also show that the QLIKE inflation of HAR-RS and HARQ is concentrated in a subset of assets: median QLIKE for HARQ drops from 16.46 (mean) to 11.12 (median) at $h = 1$, indicating that the inflation is widespread but reduced in magnitude.

Levels-based HAR extensions are unstable in this evaluation. Unconstrained OLS estimation of HAR variants can produce negative or near-zero forecasts. HAR-RS and HARQ produce QLIKE values of 3.73 and 16.46 at $h = 1$, respectively, rising to 9.40 and 12.42 at $h = 22$. These inflated values arise because the OLS-estimated models occasionally produce near-zero forecasts for RV_{t+h}, which are floored at 10^{-6} for QLIKE evaluation. Across the full sample, no forecast is literally negative, but the floor binds for a small fraction of observations. Since QLIKE $= RV/\hat{RV} - \log(RV/\hat{RV}) - 1$, a forecast near zero generates an extremely large loss whenever the realized value is far from zero, and a handful of such observations can dominate the cross-sectional average. Across the equity sample, the 10^{-6} floor binds for fewer than 0.1% of TSFM forecast–date pairs but for 2–5% of HAR-RS and HARQ pairs, explaining their inflated QLIKE. The median results in Table 7 confirm that the qualitative ranking of models is unchanged. Log-HAR and ARFIMA avoid this problem entirely by operating in log space, where forecasts are positive by construction. TSFMs also produce strictly positive forecasts, since their training distributions are bounded below. This practical advantage may be as important as any differences in statistical accuracy: a model that occasionally produces degenerate forecasts is difficult to deploy in production risk systems.

Figure 2 provides a distributional view of these results, plotting QLIKE ratios (model / HAR) across all 50 assets for six models. Values below one indicate the model outperforms HAR. At $h = 1$, Log-HAR and Moirai 2.0 have the lowest median ratios (both 0.94) and

the tightest interquartile ranges, while Lag-Llama's median ratio exceeds 1.7. At $h = 5$, Moirai 2.0 pulls ahead (median 0.80, 80% of assets below one). By $h = 22$, all models improve substantially relative to HAR, with Moirai 2.0 maintaining the lowest median ratio (0.09). The box plots also reveal that TSFM performance is more dispersed than Log-HAR's at short horizons: the interquartile range for Chronos-Bolt at $h = 1$ spans 0.80–1.04, compared to 0.72–0.97 for Log-HAR, indicating that TSFMs are more variable across assets despite achieving comparable median accuracy.

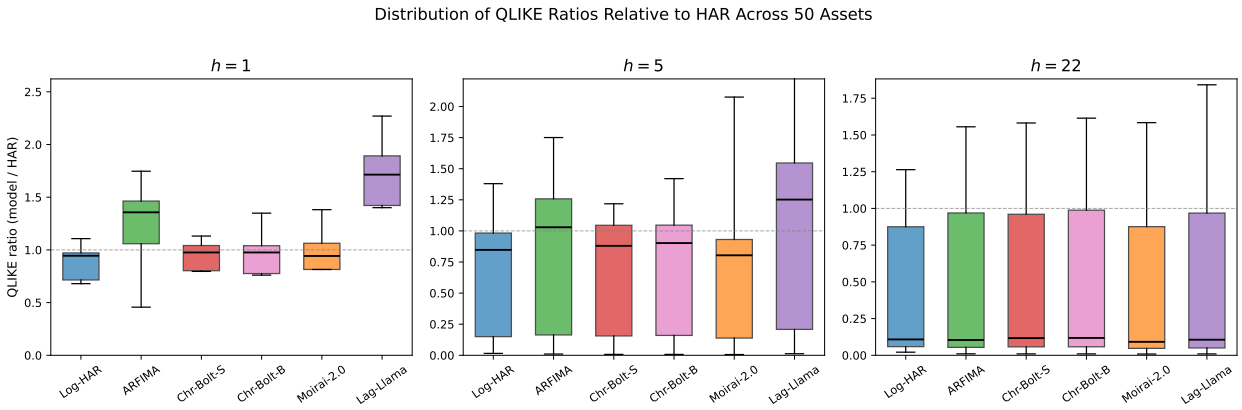

Figure 2: Distribution of QLIKE ratios (model / HAR) across 50 assets. Values below one indicate the model outperforms HAR. Boxes span the interquartile range, whiskers extend to 1.5× IQR, and outliers are suppressed for clarity. The dashed line marks parity with HAR.

5.2 Cross-Asset and Cross-Market Heterogeneity

Tables 8 and 9 extend the analysis to foreign exchange and futures markets, revealing cross-asset-class heterogeneity in model rankings.

Table 8: Forecast accuracy for 5 FX pairs (VOLARE). Values are averages across the five currency pairs. Format as Table 6.

Model	MSE ($\times 10^{-8}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.174	0.14	0.177	0.241
HAR-J	0.170	0.14	0.196	0.258
HAR-RS	0.192	0.15	0.249	0.180
HARQ	0.196	0.14	0.628	0.167
Log-HAR	0.174	0.13	**0.163**	0.247
ARFIMA	0.207	0.16	0.242	0.107
Chronos-Bolt-S	0.168	0.12	0.174	0.277
Chronos-Bolt-B	**0.167**	**0.12**	0.176	**0.279**
Moirai-2.0-S	0.222	0.19	0.329	-0.008
Lag-Llama	0.207	0.17	0.305	0.080
Panel: $h = 5$ (5 days)				
HAR	0.082	0.12	0.143	0.284
HAR-J	0.079	0.12	0.149	0.305
HAR-RS	0.098	0.13	0.240	0.173
HARQ	0.085	0.12	0.399	0.259
Log-HAR	0.080	0.11	**0.120**	0.306
ARFIMA	0.098	0.13	0.178	0.147
Chronos-Bolt-S	**0.075**	**0.10**	0.128	**0.353**
Chronos-Bolt-B	0.075	0.10	0.129	0.353
Moirai-2.0-S	0.110	0.17	0.250	-0.025
Lag-Llama	0.098	0.14	0.194	0.117
Panel: $h = 22$ (22 days)				
HAR	0.052	0.13	0.186	0.171
HAR-J	0.052	0.12	0.197	0.177
HAR-RS	0.065	0.13	0.223	0.004
HARQ	0.053	0.12	0.367	0.155
Log-HAR	0.049	0.12	**0.147**	0.205
ARFIMA	0.060	0.12	0.185	0.033
Chronos-Bolt-S	**0.047**	**0.10**	0.161	**0.250**
Chronos-Bolt-B	0.047	0.10	0.163	0.248
Moirai-2.0-S	0.064	0.16	0.212	-0.093
Lag-Llama	0.053	0.13	0.164	0.119

Table 9: Forecast accuracy for 5 futures contracts (VOLARE). Values are averages across the five contracts. Format as Table 6.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	14.818	2.72	0.225	0.244
HAR-J	14.958	2.65	0.707	**0.258**
HAR-RS	15.503	2.81	$5.82^{\dagger}$	0.180
HARQ	15.156	2.65	$22.34^{\dagger}$	0.089
Log-HAR	**14.143**	2.09	**0.193**	0.225
ARFIMA	16.437	2.95	0.346	0.009
Chronos-Bolt-S	14.335	1.99	0.213	0.237
Chronos-Bolt-B	14.251	**1.95**	0.221	0.246
Moirai-2.0-S	14.509	1.99	0.289	0.213
Lag-Llama	14.860	2.77	$1.12^{\dagger}$	0.074
Panel: $h = 5$ (5 days)				
HAR	4.574	2.56	0.434	0.253
HAR-J	4.918	2.48	0.839	0.245
HAR-RS	4.615	2.60	$4.07^{\dagger}$	0.161
HARQ	4.792	2.49	$22.10^{\dagger}$	0.181
Log-HAR	4.442	1.98	**0.165**	0.226
ARFIMA	6.262	2.70	0.269	-0.040
Chronos-Bolt-S	**4.352**	1.80	0.185	0.273
Chronos-Bolt-B	4.384	**1.77**	0.196	**0.275**
Moirai-2.0-S	4.523	1.80	0.207	0.246
Lag-Llama	4.859	2.48	0.333	0.072
Panel: $h = 22$ (22 days)				
HAR	2.169	2.77	$1.08^{\dagger}$	0.078
HAR-J	2.234	2.73	$1.23^{\dagger}$	0.071
HAR-RS	2.201	2.89	$4.01^{\dagger}$	-0.034
HARQ	2.126	2.66	$17.88^{\dagger}$	-0.051
Log-HAR	2.001	2.16	0.280	0.130
ARFIMA	3.681	2.65	0.286	-0.296
Chronos-Bolt-S	1.873	1.92	0.282	0.168
Chronos-Bolt-B	1.886	1.87	0.286	0.167
Moirai-2.0-S	**1.855**	**1.83**	**0.238**	**0.201**
Lag-Llama	2.062	2.29	0.336	0.082

Foreign exchange. The FX sample covers five major currency pairs with lower volatility levels ($\bar{RV} \approx 3 \times 10^{-5}$) and moderate persistence ($\bar{\rho}_1 = 0.52$). For FX, Chronos-Bolt consistently outperforms Moirai 2.0. At $h = 1$, Log-HAR achieves the lowest QLIKE (0.163),

with Chronos-Bolt-Small close behind (0.174), while Moirai 2.0 underperforms substantially (QLIKE = 0.329, $R^2_{\text{OOS}} = -0.008$). At $h = 5$, Log-HAR retains the QLIKE lead (0.120 vs. 0.128 for Chronos-Bolt-Small), but Chronos-Bolt achieves the best MSE and R^2_{OOS} (0.353). At $h = 22$, Log-HAR again leads on QLIKE (0.147), while Chronos-Bolt achieves the best R^2_{OOS} (0.250). Moirai 2.0's poor FX performance ($R^2_{\text{OOS}} < 0$ at all horizons) is not driven by a single currency pair: per-asset QLIKE analysis confirms that Moirai underperforms all competing models on every FX pair at every horizon. This systematic weakness stands in contrast to its equity dominance and suggests that Moirai's training data or architecture is less suited to the smoother, lower-amplitude dynamics of currency volatility.

Futures. The futures sample displays the widest cross-asset heterogeneity in volatility dynamics. The E-mini S&P 500 (ES) has first-order autocorrelation of 0.779, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$). At $h = 1$, Log-HAR achieves the lowest QLIKE (0.193), followed by Chronos-Bolt-Small (0.213) and HAR (0.225). Moirai 2.0 ranks behind (0.289). At $h = 5$, Log-HAR retains the QLIKE lead (0.165), with Chronos-Bolt-Small at 0.185 and Chronos-Bolt-Base at 0.196. The TSFMs achieve the best MSE and R^2_{OOS}. At the monthly horizon, Moirai 2.0 recovers and achieves the lowest QLIKE (0.238), MSE, and MAE, while levels-based HAR variants produce inflated QLIKE (HAR: 1.08, HARQ: 17.88).

Cross-asset patterns. Three patterns emerge from the cross-asset comparison. First, Log-HAR is consistently the best or near-best econometric specification across all asset classes, confirming the practical value of the log transformation for ensuring forecast positivity and aligning with QLIKE's loss structure. Second, the relative ranking of TSFMs shifts across asset classes: Moirai 2.0 dominates equities (where persistence is high), while Chronos-Bolt performs better for FX and short-horizon futures (where dynamics are smoother or less persistent). Third, all levels-based HAR extensions produce inflated QLIKE for futures, where extreme volatility events in Natural Gas and Crude Oil expose the negative-forecast

problem. These asset-class-dependent results highlight the importance of multi-market evaluations: a model that appears dominant on equities alone may not generalize to other asset classes.

5.3 Drivers of TSFM Forecast Performance

The cross-asset results above show that TSFM rankings vary across asset classes, raising the question of what asset-level characteristics drive relative forecast performance. Carriero et al. (2024) find that TSFMs struggle with highly persistent macroeconomic series (first-order partial autocorrelation above 0.7). We test whether a similar pattern holds for realized volatility by examining the relationship between each asset's first-order autocorrelation ρ_1 and the TSFM's QLIKE ratio relative to HAR.

Figure 3 plots QLIKE ratios (model / HAR) at $h = 1$ against ρ_1 for 50 assets, separately for each TSFM. Unlike the macroeconomic setting, we find no statistically significant relationship between persistence and TSFM relative performance: the rank correlation is $r = 0.06$ ($p = 0.68$) for Chronos-Bolt, $r = 0.02$ ($p = 0.88$) for Moirai 2.0, and $r = -0.17$ ($p = 0.23$) for Lag-Llama. Splitting assets at the median ρ_1 confirms this: Chronos-Bolt beats HAR for 68% of high-persistence assets and 52% of low-persistence assets, while Moirai 2.0 beats HAR for 64% in both groups.

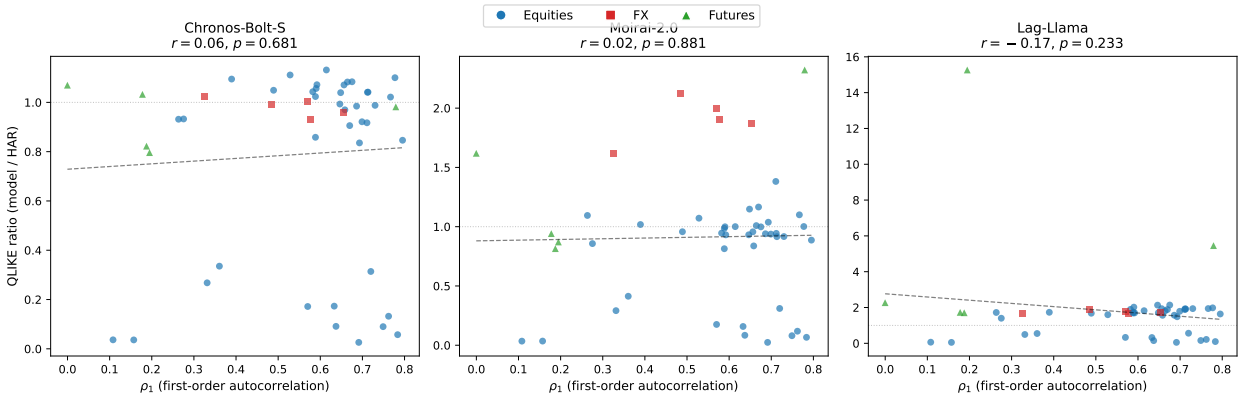

Figure 3: QLIKE ratio (model / HAR) at $h = 1$ vs. first-order autocorrelation ρ_1 for 50 assets. Values below one indicate the TSFM outperforms HAR. Each point is one asset, colored by class (blue: equities, red: FX, green: futures). Dashed line: OLS trend; r and p are the Pearson correlation and its p-value.

This null result is informative. Realized volatility, unlike many macroeconomic series, is stationary and mean-reverting by construction. Even the least persistent assets in our sample (Gold, $\rho_1 \approx 0$) exhibit the positive, right-skewed distribution that TSFMs encounter frequently in their pretraining corpora. The variation in TSFM performance across assets is better explained by asset class than by persistence: Moirai 2.0 dominates equities regardless of individual ρ_1, while Chronos-Bolt dominates FX and futures regardless of persistence. This suggests that the relevant driver is the match between each TSFM's implicit prior (shaped by its training data composition) and the distributional characteristics of each asset class, rather than the degree of serial dependence.

6 Statistical Significance and Model Confidence Sets

Having established the aggregate performance rankings, we now test the statistical significance of these differences. We apply four complementary tools: MCS to identify the subset of models that cannot be distinguished from the best, pairwise DM tests to quantify directional win rates, MZ regressions to assess forecast efficiency, and GR fluctuation tests to

detect time variation in relative performance.

Table 10 reports MCS inclusion rates for 40 equities, Figure 4 visualizes the inclusion patterns, and Table 11 summarizes pairwise DM test results across all 50 assets.

Table 10: Model Confidence Set inclusion rates for 40 U.S. equities (VOLARE). Each cell reports the percentage of stocks for which the model is included in the MCS at the 10% significance level (QLIKE loss, $T_{\max}$ statistic, block bootstrap with $B = 10{,}000$).

Model	$h = 1$ (%)	$h = 5$ (%)	$h = 22$ (%)
HAR	42.5	42.5	20.0
HAR-J	17.5	35.0	20.0
HAR-RS	2.5	2.5	12.5
HARQ	0.0	0.0	7.5
Log-HAR	90.0	65.0	75.0
ARFIMA	2.5	30.0	87.5
Chronos-Bolt-S	42.5	30.0	37.5
Chronos-Bolt-B	37.5	25.0	30.0
Moirai-2.0-S	77.5	100.0	97.5
Lag-Llama	0.0	5.0	95.0

Table 11: Diebold–Mariano test: pairwise win rates. Each cell reports the percentage of pairwise comparisons (across 50 assets $\times$ 9 opponents) in which the row model achieves significantly lower QLIKE at the 5% level.

Model	$h = 1$	$h = 5$	$h = 22$
HAR	37.6%	20.4%	5.1%
HAR-J	22.7%	18.2%	4.9%
HAR-RS	12.4%	3.1%	0.7%
HARQ	0.7%	1.1%	0.2%
Log-HAR	64.7%	38.0%	16.0%
ARFIMA	28.9%	15.6%	11.6%
Chronos-Bolt-S	48.2%	30.9%	9.8%
Chronos-Bolt-B	47.3%	27.1%	8.0%
Moirai-2.0-S	53.1%	49.1%	22.0%
Lag-Llama	15.8%	12.7%	14.4%

Each cell reports the percentage of pairwise DM tests (across 50 assets $\times$ 9 opponents = 450 tests) in which the row model has significantly lower QLIKE at the 5% level.

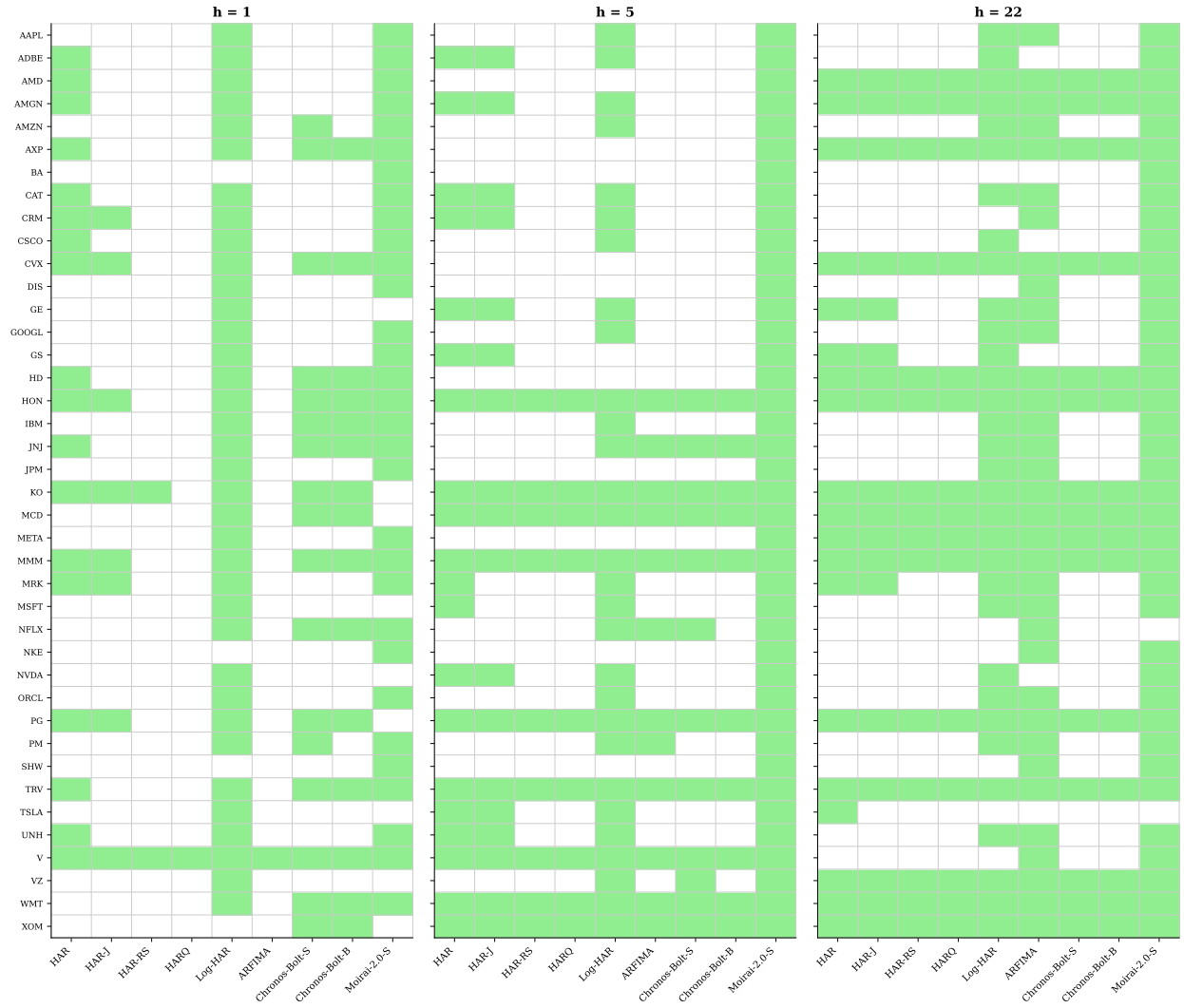

Figure 4: Model Confidence Set inclusion for 40 U.S. equities. Green cells indicate that the model is included in the MCS at the 10% significance level (QLIKE loss). Left: $h = 1$; center: $h = 5$; right: $h = 22$.

MCS results for equities. At $h = 1$, Log-HAR enters the MCS for 90% of assets, confirming its status as the single most reliable specification at the daily horizon. Moirai 2.0 enters for 77.5%, Chronos-Bolt-Small and HAR each for 42.5%. Lag-Llama enters for 0%, indicating that it is systematically outperformed at short horizons. Levels-based HAR variants are also largely excluded: HAR-RS enters for 2.5% and HARQ for 0%.

At $h = 5$, the picture shifts. Moirai 2.0 enters the MCS for 100% of assets, followed by

Log-HAR (65%), HAR (42.5%), and Chronos-Bolt-Small (30%). Lag-Llama enters for only 5%.

At $h = 22$, the ranking changes markedly. Moirai 2.0 enters for 97.5%, Lag-Llama for 95%, ARFIMA for 87.5%, Log-HAR for 75%, and Chronos-Bolt-Small for 37.5%. Lag-Llama's sharp improvement from 0% at $h = 1$ to 95% at $h = 22$ suggests that its lag-based tokenization captures long-range dependencies more effectively than short-term dynamics. The high ARFIMA inclusion rate at $h = 22$ is consistent with its long-memory parameterization: at the monthly horizon, fractional differencing captures slow mean reversion that the HAR's three fixed lags approximate less precisely.

DM pairwise tests. The DM test is computed pairwise: for each of the 10 models, we test it against each of the remaining 9 models on each of the 50 assets, yielding $10 \times 9 \times 50 = 4{,}500$ individual tests. Table 11 reports the fraction of pairwise DM tests in which each model achieves significantly lower QLIKE at the 5% level, aggregated across 50 assets and 9 opponents. At $h = 1$, Log-HAR wins the highest fraction of pairwise comparisons (64.7%), followed by Moirai 2.0 (53.1%), Chronos-Bolt-Small (48.2%), and HAR (37.6%). The TSFMs' strong showing on this directional metric (where each win requires both statistical significance and lower loss) confirms that their QLIKE advantage over levels-based HAR variants is not driven by a few outlier assets. At $h = 5$ and $h = 22$, Moirai 2.0 achieves the highest win rate (49.1% and 22.0%, respectively), reflecting its broad dominance at longer horizons. Lag-Llama's win rate rises from 15.8% at $h = 1$ to 14.4% at $h = 22$, consistent with its improved MCS inclusion at longer horizons. Figure 5 plots the cumulative QLIKE differential between Log-HAR and Moirai 2.0 at $h = 1$, showing that Moirai's advantage accumulates steadily after the COVID-19 onset.

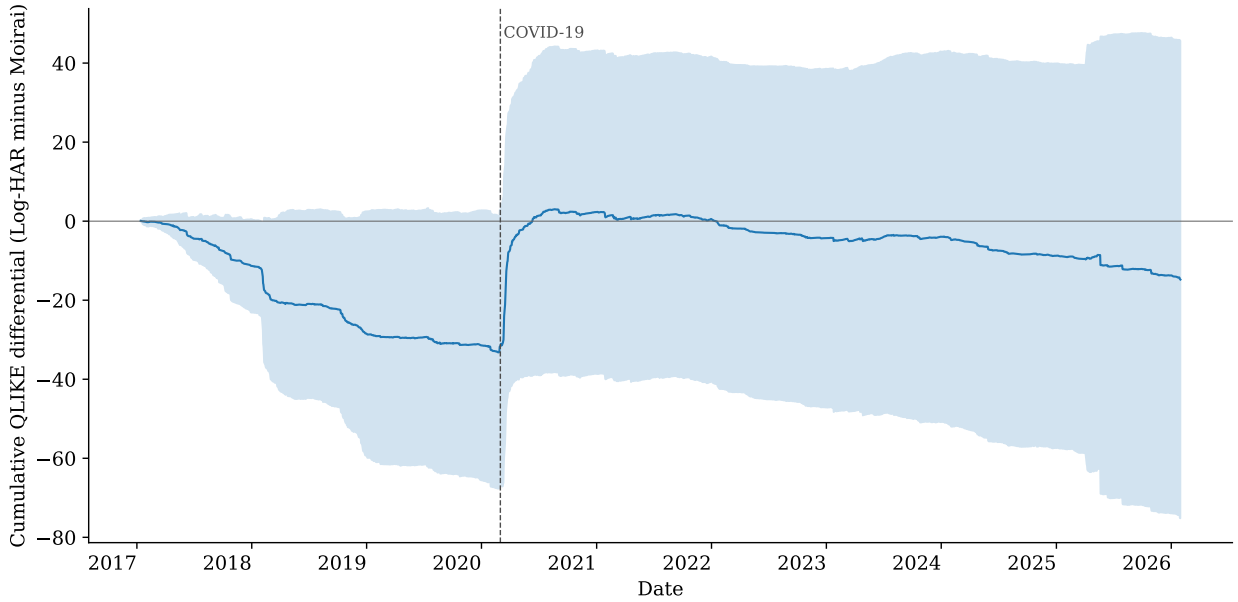

Figure 5: Cumulative QLIKE loss differential (Log-HAR minus Moirai 2.0) at $h = 1$, averaged across 40 equities. Positive values indicate that Moirai 2.0 outperforms Log-HAR. The shaded area is ± 1 cross-sectional standard deviation. The vertical dashed line marks March 2020 (COVID-19 onset).

Forecast efficiency: MZ regressions. Table 12 reports average MZ regression results across 50 assets. The MZ regression, $RV_t = \alpha + \beta \hat{f}_t + \varepsilon_t$, tests whether forecasts are efficient: under the null, $\alpha = 0$ and $\beta = 1$.

At $h = 1$, Moirai 2.0 and ARFIMA produce the most well-calibrated forecasts ($\hat{\beta} = 1.05$ for both), closest to the theoretical ideal of unity. The HAR family overshoots modestly ($\hat{\beta} = 1.29$ for HAR, 1.25 for HAR-J), while Log-HAR overshoots substantially ($\hat{\beta} = 1.77$), reflecting the bias-correction step that converts log-space forecasts back to levels. Among TSFMs, Chronos-Bolt achieves $\hat{\beta} = 1.20$ and the highest MZ R^2 (0.40–0.42), indicating that its forecasts explain a larger fraction of the variation in realized values than any econometric benchmark. The joint efficiency null ($\alpha = 0$, $\beta = 1$) is rejected for 98–100% of assets for all TSFMs, compared to 54–58% for HAR and HAR-J. This seemingly paradoxical result (TSFMs are more accurate on MSE and R^2 but more frequently reject efficiency) arises

because the MZ test has greater power when R^2 is high. A systematic $\hat{\beta} > 1$ indicates that TSFMs' forecast variation is slightly compressed relative to realized variation, a pattern consistent with the shrinkage implicit in their pretrained priors.

At $h = 22$ (Table 21), all econometric models exhibit substantial undershoot ($\hat{\beta} = 0.30$–0.58), with MZ R^2 falling to 0.07–0.14. TSFMs retain substantially better calibration: Moirai 2.0 achieves $\hat{\beta} = 0.99$ and the highest R^2 (0.19), while Chronos-Bolt-Small achieves $\hat{\beta} = 1.12$. This confirms that TSFMs' advantage at longer horizons is not only in loss-function terms but also in the structural quality of their forecasts.

Model	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	p(joint)	% Reject
Econometric benchmarks						
HAR	−0.0000	1.288	0.365	20.69	0.154	58%
HAR-J	−0.0000	1.245	0.374	11.02	0.146	54%
HAR-RS	0.0000	1.123	0.309	24.11	0.099	72%
HARQ	0.0001	0.893	0.142	13.52	0.028	82%
Log-HAR	−0.0001	1.770	0.308	8.25	0.022	90%
ARFIMA	0.0000	1.054	0.171	18.93	0.059	84%
Foundation models						
Chronos-Bolt-S	0.0000	1.200	0.402	34.50	0.004	98%
Chronos-Bolt-B	0.0000	1.253	0.423	42.74	0.003	98%
Moirai-2.0-S	0.0000	1.050	0.378	22.17	0.002	98%
Lag-Llama	−0.0001	1.499	0.147	7.21	0.041	76%

Table 12: Mincer–Zarnowitz regression ($h = 1$)

Mincer–Zarnowitz forecast efficiency regression, $h = 1$. Cross-sectional averages across 50 assets (40 equities, 5 FX, 5 futures). The regression is $RV_t = \alpha + \beta \hat{f}_t + \varepsilon_t$; under forecast efficiency, $\alpha = 0$ and $\beta = 1$. The F-statistic tests the joint null with HAC (Newey–West) standard errors. % Reject reports the fraction of assets for which the joint null is rejected at the 5% level.

Time-varying performance: GR fluctuation test. Figure 6 plots the rolling DM statistic (QLIKE loss) of each model against HAR at $h = 1$, averaged across 50 assets. A negative rolling DM indicates that the comparison model produces lower QLIKE than HAR during that window; values beyond the ±2.80 critical-value bands indicate rejection of the equal predictive ability null at the 5% level.

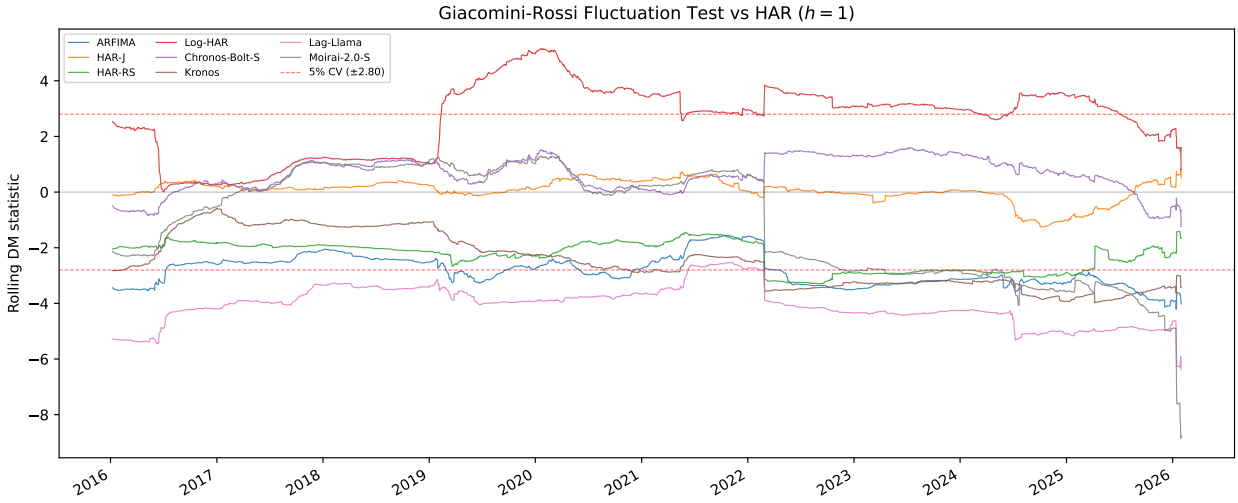

Figure 6: GR fluctuation test: rolling DM statistic (QLIKE loss) of each model vs. HAR at $h = 1$, averaged across 50 assets. Negative values indicate the comparison model outperforms HAR. Dashed red lines mark the 5% critical values (±2.80). The window size is 30% of the sample.

Three patterns emerge. First, the relative performance of all models against HAR is time-varying: the null of constant relative predictive ability is rejected for 100% of assets for Log-HAR, ARFIMA, and Lag-Llama, and for 62–98% for the other TSFMs. This means that unconditional DM tests and MCS results, while informative about average performance, mask substantial temporal variation. Second, Log-HAR's advantage over HAR is concentrated in the 2019–2021 period, where it crosses above the upper critical value, indicating that it significantly *underperforms* HAR during the COVID-induced volatility spike. This is consistent with the log-space bias correction amplifying forecast errors during regime changes. After 2022, Log-HAR's advantage narrows and approaches zero. Third, TSFMs (which enter the sample around 2022 when sufficient context becomes available) immediately settle below the lower critical-value band, indicating persistent and significant outperformance of HAR. Moirai 2.0 shows the most stable advantage.

Figures 7 and 8 present the corresponding plots at $h = 5$ and $h = 22$. At $h = 22$, the fluctuation test rejects less frequently (24–78% across models), consistent with the well-

known finding that forecast differences diminish at longer horizons.

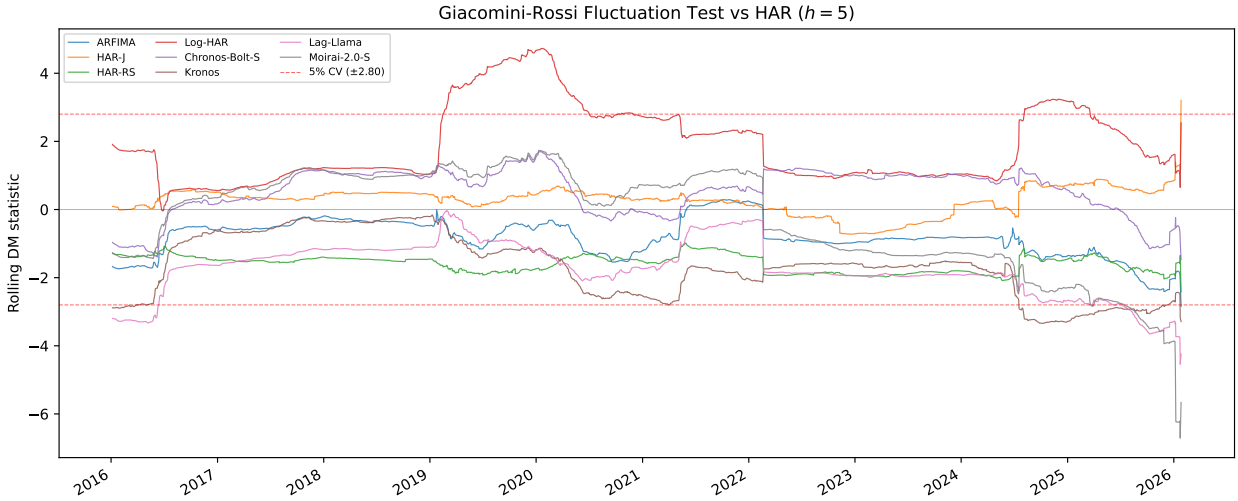

Figure 7: GR fluctuation test at $h = 5$. Each panel plots the rescaled cumulative loss differential between the column model and HAR, with ±5% critical values (dashed).

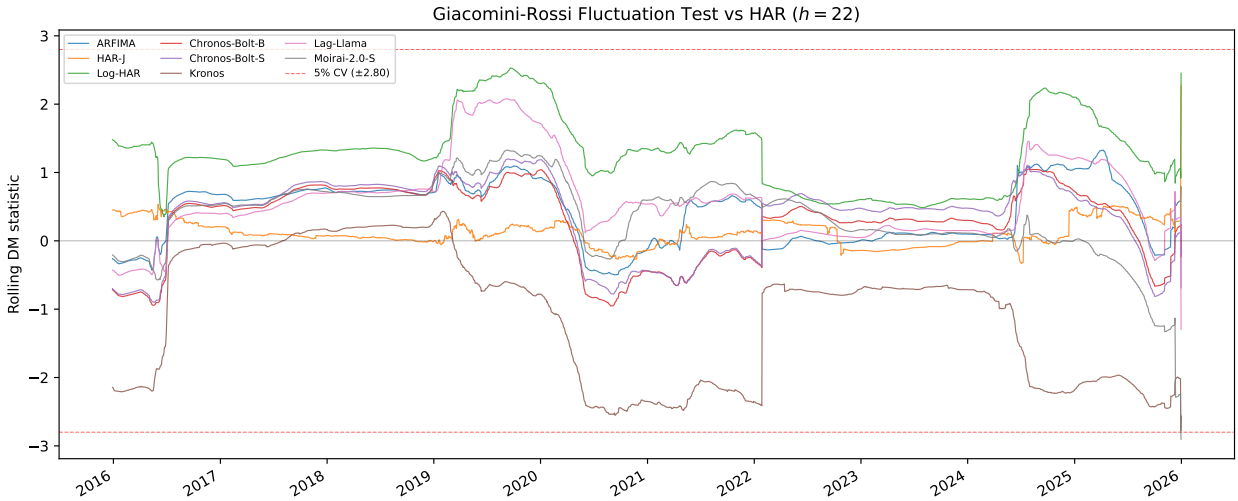

Figure 8: GR fluctuation test at $h = 22$. Each panel plots the rescaled cumulative loss differential between the column model and HAR, with ±5% critical values (dashed).

Interpretation. The MCS, DM, MZ, and fluctuation test results jointly provide a multidimensional picture of relative forecast performance. The MCS and DM tests establish that the best zero-shot TSFMs have achieved statistical parity with the best econometric

benchmarks. The MZ regressions reveal that TSFMs produce better-calibrated forecasts at long horizons ($\hat{\beta}$ closer to 1, higher R^2), while econometric models suffer severe undershoot at $h = 22$. The GR test shows that relative performance is not constant: Log-HAR's QLIKE advantage at $h = 1$ erodes over time, and TSFMs consistently outperform HAR from their entry into the sample onward.

At $h = 1$, Log-HAR remains the marginal leader on QLIKE, but Moirai 2.0 and Chronos-Bolt are statistically indistinguishable from it for most assets. At $h = 5$ and $h = 22$, Moirai 2.0 is the single model most consistently in the confidence set. However, the results also reveal heterogeneity across TSFMs: Lag-Llama is systematically excluded at short horizons, recovering only at $h = 22$. This heterogeneity suggests that architecture and pre-training data matter as much as the zero-shot paradigm itself. The practical implication is that TSFMs offer a viable alternative to HAR-family models, particularly for horizons beyond one day, but practitioners should evaluate multiple TSFM architectures rather than assuming uniform performance.

7 Realized Covariance Forecasting and Portfolio Application

The preceding analysis focused on univariate realized variance forecasting. We now extend to the multivariate setting, examining whether TSFM forecasts of realized covariance matrices can improve portfolio allocation. VOLARE provides daily realized covariance matrices for each asset class (stocks, FX, futures), computed from 5-minute intraday returns using the same methodology as the univariate realized variance measures described in Section 3. These covariance matrices serve as both the forecasting target and the ex-post evaluation benchmark.

The global minimum variance (GMV) portfolio solves the optimization problem

$$\mathbf{w}_t^{\text{GMV}} = \arg\min_{\mathbf{w}} \ \mathbf{w}'\hat{\Sigma}_{t+h|t}\,\mathbf{w} \quad \text{s.t.} \quad \mathbf{w}'\mathbf{1} = 1, \tag{10}$$

where $\hat{\Sigma}_{t+h|t}$ is the forecast covariance matrix at horizon h formed at time t, and $\mathbf{w}$ is the vector of portfolio weights constrained to sum to one (short-selling is permitted). The closed-form solution is $\mathbf{w}_t^{\text{GMV}} = \hat{\Sigma}_{t+h|t}^{-1}\mathbf{1}/(\mathbf{1}'\hat{\Sigma}_{t+h|t}^{-1}\mathbf{1})$. The GMV portfolio requires no expected return inputs and depends entirely on the covariance forecast, making it the natural vehicle for evaluating forecast quality: differences in realized portfolio variance map directly to differences in covariance forecast accuracy.

Covariance forecasting methods. We consider four covariance forecasting approaches. (i) *Element-HAR* (Chiriac and Voev, 2011): a separate HAR model is fitted to each element of the realized covariance matrix, treating variances and covariances as independent univariate series. (ii) *HAR-DRD*: the realized covariance matrix is decomposed as $\Sigma_t = D_t R_t D_t$, where D_t is the diagonal matrix of realized volatilities and R_t is the realized correlation matrix; HAR models are fitted separately to each element of D_t and R_t, and the covariance forecast is reconstructed. This DRD decomposition ensures positive semi-definiteness by construction when the forecast correlation matrix is adjusted to be positive semi-definite (PSD). (iii–iv) *Chronos-Bolt-Small* and *Moirai 2.0*: zero-shot element-wise TSFM forecasts applied to each covariance pair, followed by PSD projection. We also include the $1/N$ equal-weight portfolio as a naive benchmark.

Table 13 reports the average realized variance of GMV portfolios constructed from each forecasting model, along with portfolio turnover and concentration metrics. We present results in the same order as the univariate analysis: equities, FX, and futures.

Stocks GMV results. For the 40-stock equity portfolio, the dimensionality problem becomes acute. Only HAR-DRD produces sensible portfolios, achieving a realized variance of

5.11×10^{-5} at $h = 1$ (a 29.4% reduction relative to $1/N$, 7.24×10^{-5}). All element-wise approaches (Element-HAR, Chronos-Bolt, Moirai 2.0) fail catastrophically: their realized variances exceed $1/N$ by two to three orders of magnitude, with average daily turnover above 7 (Chronos-Bolt) and 50 (Moirai). The problem is that element-wise forecasting of 820 unique covariance pairs produces assembled matrices with extremely small eigenvalues. PSD projection with a floor of 10^{-10} yields near-singular matrices whose inverse concentrates portfolio weight on a few assets. The DRD decomposition avoids this by separately modeling volatilities and correlations, ensuring that the reconstructed matrix inherits the PSD structure of the correlation forecast. This result highlights a practical limitation: element-wise TSFM covariance forecasting does not scale to high-dimensional portfolios without structural constraints. To verify that this failure is not an artifact of the eigenvalue floor, we re-projected all stocks forecast matrices using a floor of 10^{-6} (six orders of magnitude larger than the default 10^{-10}) and re-evaluated the GMV portfolios. The higher floor reduces turnover substantially (from 13.7 to 2.2 for Chronos-Bolt and from 52.7 to 7.2 for Moirai at $h = 1$) but realized variances remain 5–7 times above $1/N$ (3.6×10^{-4} and 5.0×10^{-4} vs. 7.2×10^{-5}). HAR-DRD continues to outperform $1/N$ (4.6×10^{-5}). The failure of element-wise approaches thus reflects a fundamental structural problem, not a tuning parameter.

FX GMV results. For the 5-asset FX portfolio, all model-based approaches achieve substantial variance reduction relative to $1/N$. At $h = 1$, Chronos-Bolt-Small achieves the lowest average realized variance (3.01×10^{-6}), a 44.9% reduction relative to $1/N$ (5.46×10^{-6}). HAR-DRD is close behind (3.11×10^{-6}, 43.0% reduction), while Element-HAR (3.52×10^{-6}, 35.5%) and Moirai 2.0 (3.55×10^{-6}, 35.0%) are somewhat less effective. At $h = 5$, Chronos-Bolt and HAR-DRD maintain their advantage (45.5% and 44.7% reduction, respectively), while Moirai 2.0 produces an unstable covariance forecast that results in higher realized variance than $1/N$. At $h = 22$, all methods except Moirai 2.0 produce comparable variance reductions (31–39%).

Futures GMV results. For the 5-contract futures portfolio, Chronos-Bolt-Small achieves the lowest realized variance at $h = 1$ (3.36×10^{-5}, a 68.8% reduction vs. $1/N$) and $h = 5$ (3.55×10^{-5}, 67.2%). Moirai 2.0 performs worse, particularly at $h = 5$, where its realized variance exceeds $1/N$ by a factor of 3.5. This failure stems from Moirai's element-wise covariance forecasts being internally inconsistent: the assembled matrices are not positive semi-definite, and the eigenvalue floor used in PSD projection (10^{-10}) is too small relative to the matrix scale (typical eigenvalues $\sim 10^{-5}$). The resulting near-singular matrices produce extreme portfolio weights on a handful of dates, which dominate the average realized variance. Chronos-Bolt does not suffer this problem because its forecasts produce matrices with minimum eigenvalues around 10^{-6}, keeping condition numbers below 10.

Turnover. The $1/N$ portfolio has zero turnover by construction. Among model-based portfolios in the low-dimensional FX and futures settings, HAR-DRD achieves the lowest turnover (0.080 for FX at $h = 1$), followed by Moirai 2.0 (0.060) and Chronos-Bolt (0.116). The DRD decomposition's smoothness advantage comes from its separate modeling of volatilities and correlations, which dampens spurious variation in off-diagonal elements.

Covariance forecast accuracy. Table 14 reports element-wise forecast accuracy across all three asset classes. Two metrics are computed: the normalized Frobenius norm of the forecast error matrix ($\|\hat{\Sigma}_t - \Sigma_t\|_F/N$) and the QLIKE loss on diagonal (variance) elements. The results show that TSFMs produce competitive covariance forecasts at the element level, even when the assembled matrices are ill-conditioned for portfolio optimization. For FX at $h = 1$, Chronos-Bolt achieves the lowest Frobenius norm (0.86×10^{-5}) and near-best diagonal QLIKE (0.145 vs. 0.133 for HAR-DRD). For futures, Chronos-Bolt and HAR-DRD are comparable on Frobenius norm, with HAR-DRD leading on QLIKE at $h = 5$ and $h = 22$. For stocks, Chronos-Bolt achieves the lowest Frobenius norm at $h = 1$ and $h = 5$, while HAR-DRD leads on diagonal QLIKE. Moirai 2.0 is competitive on stocks (QLIKE 0.175 at $h = 1$, matching Chronos-Bolt) but weaker on FX (QLIKE 0.308). The disconnect

Model	h	Avg RV	Std RV	Turnover	Max Wt
Stocks (40 assets)					
$1/N$	1	72.38	200.30	—	—
HAR-DRD		**51.10**	124.55	0.788	0.159
Chronos-Bolt-S		$> 10^4$	—	13.695	—
Element-HAR		$> 10^4$	—	12.672	—
Moirai-2.0-S		$> 10^4$	—	52.690	—
$1/N$	5	72.39	130.35	—	—
HAR-DRD		**52.81**	81.06	0.789	0.159
Chronos-Bolt-S		$> 10^4$	—	7.639	—
Element-HAR		$> 10^4$	—	12.691	—
Moirai-2.0-S		$> 10^4$	—	48.779	—
$1/N$	22	72.21	93.25	—	—
HAR-DRD		**57.94**	80.69	0.790	0.159
Chronos-Bolt-S		$> 10^4$	—	7.710	—
Element-HAR		$> 10^4$	—	12.774	—
Moirai-2.0-S		$> 10^4$	—	51.283	—
Forex (5 assets)					
$1/N$	1	5.46	10.30	—	—
Element-HAR		3.52	6.50	0.096	0.383
HAR-DRD		3.11	4.50	0.080	0.370
Chronos-Bolt-S		**3.01**	4.78	0.116	0.369
Moirai-2.0-S		3.55	9.83	0.060	0.384
$1/N$	5	5.73	6.48	—	—
Element-HAR		3.58	5.45	0.096	0.383
HAR-DRD		3.17	2.99	0.080	0.370
Chronos-Bolt-S		**3.12**	3.69	0.090	0.369
Moirai-2.0-S		21.72	1083.82	0.104	0.397
$1/N$	22	5.43	4.32	—	—
Element-HAR		3.71	5.89	0.096	0.383
HAR-DRD		3.30	2.45	0.080	0.370
Chronos-Bolt-S		**3.29**	3.00	0.074	0.370
Moirai-2.0-S		3.52	6.06	0.046	0.382
Futures (5 contracts)					
$1/N$	1	107.77	316.84	—	—
Element-HAR		37.47	47.56	0.116	0.518
HAR-DRD		35.31	35.42	0.129	0.523
Chronos-Bolt-S		**33.63**	33.30	0.114	0.520
Moirai-2.0-S		45.08	156.03	0.264	0.552
$1/N$	5	107.88	207.81	—	—
Element-HAR		39.89	83.62	0.116	0.518
HAR-DRD		36.83	33.59	0.129	0.523
Chronos-Bolt-S		**35.46**	32.45	0.088	0.514
Moirai-2.0-S		376.26	19571.31	0.321	0.560
$1/N$	22	108.35	151.33	—	—
Element-HAR		42.66	50.69	0.116	0.518
HAR-DRD		40.75	38.02	0.129	0.523
Chronos-Bolt-S		**39.88**	44.59	0.065	0.503
Moirai-2.0-S		59.63	670.30	0.154	0.515

Table 13: Global minimum variance portfolio performance

Global minimum variance portfolio performance. Avg RV is the average daily realized variance of the GMV portfolio ($\times 10^{-6}$). Turnover is the average daily absolute weight change. For stocks, element-wise approaches (Element-HAR, Chronos-Bolt, Moirai) produce near-singular matrices that cause extreme portfolio weights; their Avg RV values are orders of magnitude above $1/N$ and are omitted from the Std RV and Max Wt columns. Bold indicates the lowest Avg RV in each panel.

	Model	$h=1$		$h=5$		$h=22$	
		Frob ($\times 10^{-5}$)	QLIKE	Frob ($\times 10^{-5}$)	QLIKE	Frob ($\times 10^{-5}$)	QLIKE
Forex	Element-HAR	1.06	0.149	0.89	0.105	0.88	0.116
	HAR-DRD	0.96	0.133	0.79	**0.092**	0.79	**0.112**
	Chronos-Bolt-S	**0.86**	**0.145**	**0.72**	0.112	**0.72**	0.151
	Moirai-2.0-S	1.38	0.308	1.28	0.245	1.17	0.211
Futures	Element-HAR	21.64	0.227	20.04	0.183	20.46	0.214
	HAR-DRD	15.06	0.158	**13.35**	**0.125**	**13.71**	**0.175**
	Chronos-Bolt-S	**14.87**	**0.173**	13.74	0.160	14.87	0.260
	Moirai-2.0-S	15.20	0.228	13.88	0.181	14.41	0.217
Stocks	Element-HAR	7.22	0.196	6.18	0.171	6.46	0.212
	HAR-DRD	6.87	0.166	5.88	**0.150**	**5.81**	**0.204**
	Chronos-Bolt-S	**6.72**	**0.176**	**5.84**	0.181	5.89	0.303
	Moirai-2.0-S	7.48	0.175	6.37	0.160	6.20	0.237

Table 14: Element-wise covariance forecast accuracy

Covariance forecast accuracy by asset class. Avg Frobenius is the average normalized Frobenius norm of the forecast error matrix ($\|\hat{\Sigma}_t - \Sigma_t\|_F/N$). Diag QLIKE is the average QLIKE computed on diagonal (variance) elements. Bold indicates the best value in each panel×horizon cell.

between element-wise accuracy and portfolio performance (Chronos-Bolt's Frobenius norm is lower than HAR-DRD's for stocks, yet its portfolios fail) confirms that the problem lies in the spectral properties of the assembled matrix, not in the accuracy of individual element forecasts. Potential solutions include applying the DRD decomposition to TSFM forecasts (forecasting volatilities and correlations separately, then reconstructing), imposing shrinkage toward a structured target (Ledoit and Wolf, 2004), or using natively multivariate TSFM architectures that produce positive semi-definite outputs by construction.

8 Robustness Checks

We assess the stability of our main findings along several dimensions.

Sub-sample stability: pre- and post-COVID. Table 15 splits the equity sample at March 1, 2020, and reports forecast accuracy separately for the pre-COVID (2015–2020) and post-COVID (2020–2026) periods. The sub-sample analysis reveals that TSFM performance is regime-robust, while levels-based econometric models are not.

In the pre-COVID period, Log-HAR achieves the lowest QLIKE at all horizons (0.206 at

$h = 1$, 0.182 at $h = 5$, 0.321 at $h = 22$), though Chronos-Bolt-Base matches it on MSE and R^2_{OOS} at $h = 1$. The pre-COVID period features lower average volatility and fewer extreme events, conditions that favor the parsimonious HAR specification.

In the post-COVID period, TSFMs dominate across all metrics and horizons. Moirai 2.0 achieves the lowest QLIKE at $h = 1$ (0.213 vs. 0.225 for Log-HAR), $h = 5$ (0.174 vs. 0.252), and $h = 22$ (0.168 vs. 0.335), while also achieving the highest R^2_{OOS} (0.430, 0.378, 0.148). The levels-based HAR variants suffer extreme QLIKE inflation in the high-volatility regime: HARQ's QLIKE rises from 2.35 pre-COVID to 23.63 post-COVID at $h = 1$, because the near-zero forecast problem is exacerbated when realized values are large. Log-HAR is more stable across regimes than its levels-based counterparts but still falls behind the TSFMs on all metrics in the post-COVID sample. The TSFM heterogeneity pattern persists across subsamples: Lag-Llama underperforms in both periods at $h = 1$ (QLIKE 0.433 pre-COVID, 0.385 post-COVID) but improves at longer horizons. These results suggest that TSFMs' advantage is concentrated in volatile market conditions, where their flexible nonparametric forecasts adapt more effectively than the HAR's linear structure.

Estimation window: 252 vs. 512 days. A potential concern is that TSFMs receive 512 days of context while econometric models use 252-day estimation windows, creating an information asymmetry that may favor TSFMs. To address this, we re-estimate all six econometric benchmarks with a 512-day rolling window, matching the TSFM context length. Table 16 reports the results at $h = 1$ for the 40-equity sample. Five of six models improve under the longer window: HAR R^2_{OOS} rises from 0.340 to 0.435, and levels-HAR QLIKE inflation decreases substantially (e.g., HAR-RS QLIKE drops from 3.48 to 0.57). The exception is ARFIMA, whose long-memory estimation degrades with a 512-day window (R^2_{OOS} falls from 0.132 to -0.332). Cross-model R^2_{OOS} rankings are highly stable (Spearman $\rho = 0.94$, $p < 0.01$). Importantly, even after granting econometric models the same 512-day information set, Chronos-Bolt and Moirai zero-shot QLIKE (0.19–0.23; Table 6) remain competitive

Model	Pre-COVID				Post-COVID			
	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel A: $h = 1$								
HAR	0.089	0.984	0.494	0.223	0.328	1.684	1.809†	0.330
HAR-J	0.090	0.976	0.395	0.204	0.324	1.667	1.539†	0.345
HAR-RS	0.102	1.037	1.312†	0.158	0.356	1.797	5.151†	0.259
HARQ	0.091	0.959	2.346†	0.175	0.406	1.780	23.625†	0.091
Log-HAR	0.087	0.862	**0.206**	0.242	0.348	1.545	0.225	0.225
ARFIMA	0.123	1.148	0.328	-0.008	0.392	1.896	0.312	0.147
Chronos-Bolt-S	0.086	0.834	0.235	0.247	0.285	1.363	0.226	0.410
Chronos-Bolt-B	0.084	**0.823**	0.234	0.275	0.281	**1.353**	0.228	0.422
Moirai-2.0-S	0.084	0.857	0.251	0.235	**0.277**	1.360	**0.213**	**0.430**
Lag-Llama	0.106	1.188	0.433	0.007	0.388	1.899	0.385	0.120
Kronos	**0.082**	0.895	0.954	**0.287**	0.309	1.527	1.166†	0.340
Panel B: $h = 5$								
HAR	0.050	0.972	0.421	0.029	0.221	1.643	3.877†	0.193
HAR-J	0.052	0.966	0.427	0.018	0.214	1.621	3.529†	0.219
HAR-RS	0.061	1.025	2.051†	-0.098	0.249	1.777	9.622†	0.077
HARQ	0.050	0.954	1.814†	0.036	0.265	1.694	23.517†	-0.008
Log-HAR	**0.044**	0.839	**0.182**	0.190	0.226	1.500	0.252	0.108
ARFIMA	0.063	1.014	0.263	-0.063	0.227	1.672	0.237	0.167
Chronos-Bolt-S	0.047	0.791	0.241	0.168	0.170	1.256	0.200	0.348
Chronos-Bolt-B	0.046	**0.782**	0.240	0.187	0.172	1.261	0.204	0.337
Moirai-2.0-S	0.045	0.789	0.217	0.202	**0.162**	**1.216**	**0.174**	**0.378**
Lag-Llama	0.056	1.024	0.274	0.026	0.246	1.638	0.321	0.020
Kronos	0.045	0.842	0.700	**0.229**	0.195	1.421	0.953	0.220
Panel C: $h = 22$								
HAR	0.095	1.145	1.587†	-0.032	0.118	1.621	8.230†	-0.070
HAR-J	0.095	1.140	1.510†	-0.028	0.117	1.608	8.084†	-0.047
HAR-RS	0.101	1.197	3.796†	-0.084	0.133	1.739	11.923†	-0.243
HARQ	0.095	1.134	2.323†	-0.039	0.139	1.637	17.488†	-0.331
Log-HAR	**0.091**	1.062	**0.321**	-0.006	0.116	1.470	0.335	-0.102
ARFIMA	0.098	1.150	0.361	-0.065	0.119	1.418	0.205	-0.031
Chronos-Bolt-S	0.115	1.095	0.533	-0.029	0.092	1.152	0.211	0.121
Chronos-Bolt-B	0.115	1.090	0.531	-0.027	0.093	1.160	0.217	0.108
Moirai-2.0-S	0.113	**1.057**	0.427	**-0.002**	**0.089**	**1.084**	**0.168**	**0.148**
Lag-Llama	0.114	1.184	0.362	-0.007	0.105	1.285	0.239	-0.012
Kronos	0.117	1.178	1.016†	-0.029	0.102	1.298	0.854	0.003

Sub-sample forecast accuracy: pre-COVID (2015–2020) and post-COVID (2020–2026) periods for 40 U.S. equities (VOLARE). Bold indicates best in each column within each panel. † denotes QLIKE values inflated by near-zero forecasts.

Table 15: Sub-sample stability: pre- and post-COVID

with or below the best 512-day econometric QLIKE (Log-HAR at 0.20), confirming that the TSFM advantage is not an artifact of the longer context window. The same pattern holds at longer horizons: Tables 22 and 23 in Appendix A confirm that the 512-day window improves all HAR variants at $h = 5$ and $h = 22$, while ARFIMA deteriorates further.

	MSE ($\times 10^8$)		QLIKE		R^2_{OOS}	
Model	252	512	252	512	252	512
HAR	23.03	**21.38**	1.2715	**0.2107**	0.340	**0.435**
HAR-J	22.78	**20.89**	1.0356	**0.3590**	0.353	**0.442**
HAR-RS	25.18	**21.92**	3.4828	**0.5729**	0.272	**0.417**
HARQ	27.71	**24.64**	14.5773	**4.8653**	0.119	**0.270**
Log-HAR	23.98	**22.46**	0.2166	**0.1954**	0.249	**0.381**
ARFIMA	**28.39**	39.98	**0.3264**	0.4786	**0.132**	-0.332

Table 16: Estimation window sensitivity: 252 vs. 512 days

Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 1$, 40 equities). Cross-asset mean metrics. Bold indicates the better window for each model.

Context window length. Table 17 examines how TSFM forecast accuracy varies with context window length (128, 256, and 512 days). Longer context generally improves Chronos-Bolt: QLIKE decreases from 0.231 (ctx=128) to 0.222 (ctx=512) at $h = 1$, with similar patterns at $h = 5$. Moirai 2.0 shows a more nuanced pattern: while longer context improves QLIKE at $h = 1$ (0.264 to 0.243), it *worsens* performance at $h = 22$ (0.239 at ctx=128 vs. 0.251 at ctx=512), suggesting that Moirai's architecture captures long-range dependencies efficiently even from shorter windows and that additional context introduces noise at the monthly horizon. Lag-Llama benefits substantially from longer context at $h = 5$ and $h = 22$ (QLIKE drops from 0.515 to 0.297 at $h = 5$), but at $h = 1$ the relationship is non-monotonic, with the best QLIKE at ctx=256 (0.380). These results confirm that the 512-day default is appropriate for Chronos-Bolt but not universally optimal across architectures and horizons.

Table 17: Context-length sensitivity of TSFM forecasts. QLIKE and R^2_{OOS} averaged across all assets, by horizon and context window length. Bold indicates the best context length for each model–horizon pair.

		QLIKE			R^2_{OOS}		
Model	h	ctx=128	ctx=256	ctx=512	ctx=128	ctx=256	ctx=512
Chronos-Bolt-S	1	0.231	0.232	**0.222**	0.337	0.330	**0.378**
	5	0.216	0.208	**0.203**	0.259	0.278	**0.345**
	22	**0.299**	0.305	0.303	-0.183	0.089	**0.109**
Chronos-Bolt-B	1	0.401	0.245	**0.224**	0.369	0.377	**0.389**
	5	0.346	**0.206**	0.206	0.329	0.317	**0.337**
	22	0.394	**0.302**	0.307	0.087	0.098	**0.102**
Moirai-2.0-S	1	0.264	0.249	**0.243**	**0.364**	0.347	0.362
	5	0.201	0.199	**0.197**	**0.358**	0.319	0.331
	22	**0.239**	0.243	0.251	**0.149**	0.112	0.098
Lag-Llama	1	0.433	**0.380**	0.464	**0.191**	0.168	0.117
	5	0.515	0.312	**0.297**	0.027	**0.061**	0.054
	22	0.587	0.332	**0.276**	-0.076	0.003	**0.027**

QLIKE floor sensitivity. Our QLIKE implementation floors forecasts at 10^{-10} to prevent division by zero. Because levels-based HAR variants occasionally produce near-zero forecasts, this floor choice could affect model rankings. Table 18 reports cross-asset mean QLIKE at $h = 1$ under four floor values (10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}). QLIKE values are identical for floors $\leq 10^{-6}$ (Spearman rank correlation $\rho = 1.0$ vs. 10^{-10}), confirming that the floor never binds at these levels. Only the aggressive 10^{-4} floor alters rankings ($\rho = 0.60$) by clipping levels-HAR forecasts that drive QLIKE inflation (e.g., HARQ drops from 16.04 to 0.38). Our default 10^{-10} floor is conservative and does not distort model comparisons.

MZ bias-corrected evaluation. A model's low loss could reflect good calibration (i.e., alignment between the scale of forecasts and realizations) rather than genuine predictive information about future volatility dynamics. The MZ regression provides a natural correction: by regressing realized values on forecasts, one obtains intercept and slope parameters that remove any linear bias. If a model's accuracy survives this correction, its forecasts contain

Model	$1e-10$	$1e-08$	$1e-06$	$1e-04$
ARFIMA	0.3218	0.3218	0.3218	0.3195
Log-HAR	0.2189	0.2189	0.2189	0.2358
HAR-J	1.11	1.11	1.11	0.2461
HARQ	16.04	16.04	16.04	0.3827
HAR	1.38	1.38	1.38	0.2565
HAR-RS	3.63	3.63	3.63	0.2838
Lag-Llama	0.4018	0.4018	0.4018	0.3639
Chronos-Bolt-S	0.2292	0.2292	0.2292	0.2353
Kronos	1.09	1.09	1.09	0.2346
Moirai-2.0-S	0.2263	0.2263	0.2263	0.2274
Chronos-Bolt-B	0.2302	0.2302	0.2302	0.2351

Table 18: QLIKE floor sensitivity analysis

QLIKE floor sensitivity: cross-asset mean QLIKE at $h = 1$ under four floor values. Rankings are stable across floors (Spearman $\rho > 0.99$).

information beyond what a simple affine rescaling could produce. To test this, we apply a recursive MZ bias correction to all TSFM forecasts at $h = 1$: at each time t, we estimate $\hat{\alpha}_t$ and $\hat{\beta}_t$ from all observations up to $t-1$, then form the corrected forecast $\hat{f}_t^{\mathrm{MZ}} = \hat{\alpha}_t + \hat{\beta}_t f_t$ using an expanding estimation window (minimum 252 days). Table 19 compares original and MZ-corrected TSFM forecasts against econometric benchmarks, evaluated on the common post-warm-up sample. MZ correction improves QLIKE for the well-calibrated Chronos-Bolt models (Chronos-Bolt-Base: $0.2307 \rightarrow 0.1918$), but worsens QLIKE for Lag-Llama, where the linear correction pushes some forecasts toward zero. MSE and R^2_{OOS} are largely unchanged, confirming that TSFM accuracy reflects genuine predictive content rather than calibration artifacts.

9 Conclusion

This paper evaluates zero-shot time series foundation models for realized volatility and covariance forecasting across 50 assets spanning U.S. equities, foreign exchange, and futures. We compare three TSFMs (Chronos-Bolt, Moirai 2.0, and Lag-Llama) against six econo-

Model	Variant	MSE ($\times 10^8$)	QLIKE	R^2_{OOS}
HAR	Original	27.07	1.52	0.333
HAR-J	Original	26.76	1.22	0.347
HAR-RS	Original	29.44	4.04	0.266
HARQ	Original	32.98	17.97	0.105
Log-HAR	Original	28.41	0.2220	0.237
ARFIMA	Original	33.34	0.3279	0.120
Chronos-Bolt-S	Original	24.04	0.2298	0.403
Chronos-Bolt-S	MZ-Corrected	27.51	0.1969	0.322
Chronos-Bolt-B	Original	23.67	0.2307	0.415
Chronos-Bolt-B	MZ-Corrected	25.40	0.1918	0.383
Moirai-2.0-S	Original	23.37	0.2245	0.421
Moirai-2.0-S	MZ-Corrected	25.56	0.2238	0.331
Lag-Llama	Original	32.36	0.4162	0.117
Lag-Llama	MZ-Corrected	32.46	7.05	0.116
Kronos	Original	25.63	0.9452	0.342
Kronos	MZ-Corrected	25.88	7.84	0.348

Table 19: Mincer–Zarnowitz bias-corrected evaluation

MZ bias-corrected TSFM evaluation (cross-asset mean, $h = 1$). Each TSFM forecast $\hat{f}_t$ is replaced by $\hat{\alpha}+\hat{\beta}\hat{f}_t$ from a rolling Mincer–Zarnowitz regression, then re-evaluated on MSE, QLIKE, and R^2_{OOS}. Econometric benchmarks are shown unchanged for comparison.

metric specifications at three forecast horizons using the VOLARE dataset, with formal statistical testing via DM tests and MCS.

Three main findings emerge. First, the best zero-shot TSFMs are competitive with the best econometric benchmarks for univariate realized variance forecasting. This contrasts with prior negative findings on daily returns and on TimesFM 1.0, and is consistent with the hypothesis that realized volatility's positive, persistent, mean-reverting dynamics align more closely with the series on which TSFMs were pretrained. However, not all TSFMs perform equally: Lag-Llama only enters the MCS at $h = 22$. Second, model rankings depend on loss function and horizon. Log-HAR achieves the lowest QLIKE at $h = 1$ for equities, while TSFMs dominate on MSE, MAE, and R^2_{OOS} at all horizons; at $h = 5$ and $h = 22$, Moirai 2.0 leads across all metrics. Third, rankings shift across asset classes: Chronos-Bolt outperforms on FX and futures, while Moirai 2.0 dominates equities.

We also evaluate realized covariance forecasting and its downstream application to global minimum variance portfolio construction. Element-wise zero-shot TSFM forecasts of realized covariance matrices achieve substantial variance reductions (35–69%) relative to equal-weight allocation in low-dimensional portfolios (5 assets), with Chronos-Bolt performing comparably to HAR-DRD baselines. However, element-wise TSFM covariance forecasting fails for the 40-stock portfolio, where the assembled matrices are ill-conditioned despite competitive element-level accuracy. Only the structurally constrained HAR-DRD decomposition produces viable high-dimensional allocations, highlighting that covariance forecasting requires structural constraints that current element-wise TSFM approaches lack.

Several limitations should be noted. We evaluate only zero-shot TSFM performance; parameter-efficient fine-tuning may yield further gains, though the adaptation mechanisms differ across architectures (Chronos-2 supports LoRA-based adaptation through its encoder-decoder structure, while Lag-Llama uses epoch-based training through its GluonTS estimator interface). Our evaluation is restricted to point forecasts; density forecast evaluation remains an open question. VOLARE provides alternative realized measures (subsampled

realized variance, realized kernels, range-based estimators) whose effect on model rankings we leave for future work. Natively multivariate TSFM architectures or structural decompositions applied to TSFM outputs may address the high-dimensional covariance forecasting limitation. While no TSFM in our study was trained on realized volatility (Section 4.3), indirect exposure to related financial series such as exchange rates and macroeconomic indicators cannot be fully ruled out as a contributing factor; future work with held-out asset classes or temporal holdouts could further address this concern.

The broader implication is that zero-shot time series foundation models have reached a level of accuracy that makes them a practical tool for volatility and covariance forecasting. For practitioners, Moirai 2.0 and Chronos-Bolt combine competitive accuracy with zero estimation overhead and immediate deployment to new assets. At the same time, the poor performance of some TSFMs at short horizons and the failure of element-wise covariance forecasting at high dimensions caution against treating TSFMs as a universal solution. Model selection and structural constraints remain essential.

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A Supplementary Tables

This appendix collects additional results referenced in the main text. Tables A, A, and A provide per-asset QLIKE breakdowns at $h = 1$, $h = 5$, and $h = 22$ across all 50 assets. Tables 20 and 21 present Mincer–Zarnowitz regressions at $h = 5$ and $h = 22$. Tables 22 and 23 extend the estimation window sensitivity analysis to $h = 5$ and $h = 22$.

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	Lag-Llama
Stocks							
AAPL	0.2179	**0.2059**	0.3497	0.2273	0.2321	0.2061	0.4127
ADBE	0.1619	**0.1605**	0.2755	0.1755	0.1789	0.1617	0.3462
AMD	0.1653	0.1638	0.2588	0.1692	0.1720	**0.1630**	0.2877
AMGN	0.2308	**0.2121**	0.3130	0.2422	0.2369	0.2210	0.3878
AMZN	13.8320	**0.4528**	0.6815	0.5050	0.5127	0.4790	0.8656
AXP	0.2137	0.2241	0.2973	0.2111	0.2106	**0.1959**	0.4140
BA	9.3426	0.2843	0.3670	0.2439	0.2447	**0.2248**	0.4943
CAT	0.1769	0.1793	0.2530	0.1895	0.1907	**0.1693**	0.3422
CRM	0.2299	**0.2170**	0.3759	0.2516	0.2520	0.2341	0.3983
CSCO	0.1911	**0.1852**	0.3172	0.2102	0.2098	0.1913	0.3785
CVX	0.1870	0.1962	0.2593	0.1583	**0.1566**	0.1658	0.3065
DIS	2.5596	0.2216	0.3013	0.2302	0.2345	**0.2058**	0.4009
GE	0.2007	**0.1949**	0.2884	0.2173	0.2185	0.2025	0.3640
GOOGL	0.3662	**0.3088**	0.4195	0.3416	0.3519	0.3140	0.5129
GS	0.1670	0.1643	0.2697	0.1739	0.1715	**0.1529**	0.3189
HD	0.1971	0.1975	0.2923	0.1957	0.1939	**0.1836**	0.4202
HON	0.2256	0.2227	0.3146	0.2078	**0.2058**	0.2115	0.4032
IBM	0.6798	**0.1981**	0.3104	0.2133	0.2157	0.2109	0.3832
JNJ	0.2310	0.2194	0.2925	0.2152	**0.2128**	0.2529	0.3978
JPM	0.1874	0.1848	0.3035	0.1950	0.1943	**0.1768**	0.3618
KO	0.1827	0.1767	0.2511	0.1675	**0.1655**	0.2524	0.3446
MCD	3.9225	0.2386	0.2976	**0.2252**	0.2259	0.2622	0.3831
META	0.2198	0.1855	0.3188	0.1887	0.1951	**0.1790**	0.3721
MMM	0.2185	**0.2121**	0.2965	0.2271	0.2280	0.2509	0.3729
MRK	0.2064	0.2000	0.2678	0.2212	0.2184	**0.1922**	0.3507
MSFT	0.2980	**0.2718**	0.4233	0.3311	0.3256	0.3193	0.4759
NFLX	10.5396	**0.3476**	0.4306	0.3809	0.3801	0.3696	0.5480
NKE	0.2170	0.2098	0.2835	0.2105	0.2125	**0.1818**	0.3396
NVDA	0.1909	**0.1827**	0.3225	0.2017	0.2062	0.1907	0.3857
ORCL	1.6483	**0.2585**	0.3809	0.2854	0.2961	0.2608	0.5327
PG	0.2001	0.1900	0.2404	0.1812	**0.1806**	0.2332	0.3771
PM	0.2589	**0.2424**	0.2899	0.2549	0.2589	0.2433	0.4061
SHW	1.5060	0.2060	0.2484	0.1991	0.2005	**0.1784**	0.3264
TRV	0.1689	**0.1655**	0.2349	0.1726	0.1691	0.1858	0.3279
TSLA	2.2254	**0.1722**	0.2929	0.2029	0.2036	0.1869	0.3343
UNH	0.2226	**0.2147**	0.3134	0.2519	0.2527	0.2228	0.4056
V	0.8008	**0.2616**	0.3140	0.2686	0.2732	0.3309	0.4384
VZ	1.1632	**0.1859**	0.2658	0.1998	0.2035	0.2040	0.3847
WMT	0.9400	0.2522	0.3115	0.2517	**0.2498**	0.2743	0.4637
XOM	0.2045	0.2264	0.2449	0.1709	**0.1678**	0.2122	0.3034
Average	1.3774	**0.2198**	0.3142	0.2292	0.2302	0.2263	0.4017
FX							
AUDUSD	0.1617	0.1524	0.2132	**0.1504**	0.1536	0.3086	0.2664
EURUSD	0.1571	**0.1491**	0.2035	0.1576	0.1594	0.3139	0.2768
GBPUSD	0.1504	**0.1376**	0.2120	0.1492	0.1487	0.3198	0.2806
USDCAD	0.1380	**0.1308**	0.1749	0.1326	0.1331	0.2587	0.2376
USDJPY	0.2753	**0.2435**	0.4040	0.2823	0.2856	0.4456	0.4621
Average	0.1765	**0.1627**	0.2415	0.1744	0.1761	0.3293	0.3047
Futures							
C	0.3153	**0.2903**	0.3282	0.3256	0.3269	0.2969	0.5447
CL	0.1998	**0.1358**	0.3310	0.1591	0.1521	0.1741	3.0510
ES	0.2286	**0.1976**	0.5399	0.2244	0.2224	0.5307	1.2473
GC	0.1648	**0.1601**	0.2879	0.1763	0.2223	0.2669	0.3741
NG	0.2176	0.1787	0.2421	0.1790	0.1802	**0.1773**	0.3715
Average	0.2252	**0.1925**	0.3458	0.2129	0.2208	0.2892	1.1177
Overall Average	1.1421	**0.2114**	0.3101	0.2221	0.2239	0.2429	0.4636

Per-asset QLIKE loss at $h = 1$. QLIKE is computed as $\text{mean}(RV_t/\hat{RV}_t - \ln(RV_t/\hat{RV}_t) - 1)$. Bold indicates the lowest (best) QLIKE in each row. Chr-Bolt-S = Chronos-Bolt-Small, Chr-Bolt-B = Chronos-Bolt-Base, Moirai-S = Moirai-2.0-Small. HAR-J, HAR-RS, and HARQ are omitted as their QLIKE values are dominated by near-zero forecast artifacts (see Section 5.1).

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	Lag-Llama
Stocks							
AAPL	0.2447	0.2027	0.2772	0.2302	0.2371	**0.1916**	0.2930
ADBE	0.1507	0.1449	0.2189	0.1633	0.1687	**0.1397**	0.2413
AMD	0.1531	0.1451	0.1850	0.1381	0.1421	**0.1328**	0.1720
AMGN	0.2248	0.2053	0.2356	0.2363	0.2345	**0.2044**	0.3123
AMZN	19.2609	0.4705	0.5729	0.5106	0.5179	**0.4695**	0.5694
AXP	0.2035	0.2807	0.2350	0.2013	0.1992	**0.1616**	0.3605
BA	31.7150	0.4817	0.3190	0.2305	0.2268	**0.1941**	0.4017
CAT	0.1398	0.1436	0.1792	0.1586	0.1626	**0.1321**	0.2276
CRM	0.2246	0.2228	0.2950	0.2551	0.2554	**0.2227**	0.2723
CSCO	0.9244	0.1683	0.2543	0.1986	0.1988	**0.1596**	0.2837
CVX	2.5162	0.2777	0.2162	0.1504	0.1523	**0.1339**	0.3232
DIS	8.5696	0.2826	0.2174	0.2182	0.2263	**0.1704**	0.3200
GE	0.1648	**0.1627**	0.2036	0.1908	0.1885	0.1653	0.2595
GOOGL	2.1229	**0.2661**	0.3177	0.3076	0.3215	0.2700	0.3457
GS	0.1475	0.1570	0.2097	0.1515	0.1544	**0.1238**	0.2465
HD	0.1990	0.2091	0.2376	0.1875	0.1869	**0.1533**	0.3327
HON	0.2820	0.3500	0.2460	0.2014	0.1968	**0.1730**	0.3479
IBM	1.6441	0.2038	0.2341	0.2044	0.2085	**0.1654**	0.2969
JNJ	0.2565	0.2141	0.2343	0.1922	**0.1909**	0.1935	0.2918
JPM	0.1760	0.1972	0.2489	0.1943	0.1959	**0.1488**	0.3191
KO	**0.1661**	0.1815	0.2126	0.1666	0.1670	0.2116	0.3091
MCD	5.7174	0.2857	0.2302	0.2107	0.2152	**0.1992**	0.3788
META	3.0672	0.1969	0.2529	0.1829	0.1842	**0.1618**	0.2561
MMM	0.6220	**0.1950**	0.2194	0.2026	0.2040	0.2049	0.2632
MRK	0.1776	0.1687	0.1848	0.1917	0.1898	**0.1464**	0.2338
MSFT	0.2953	**0.2437**	0.3282	0.2984	0.3080	0.2600	0.3749
NFLX	14.9187	0.3343	0.3298	0.3395	0.3433	**0.3230**	0.3636
NKE	0.2098	0.2113	0.2042	0.1934	0.1956	**0.1476**	0.2729
NVDA	0.1693	**0.1642**	0.2522	0.1836	0.1878	0.1700	0.2513
ORCL	4.6326	0.2788	0.3143	0.2984	0.3103	**0.2443**	0.3914
PG	0.2826	0.2035	0.1889	**0.1569**	0.1606	0.1858	0.2956
PM	0.2212	0.1903	0.2151	0.2176	0.2217	**0.1803**	0.2951
SHW	1.6041	0.2297	0.1700	0.1677	0.1711	**0.1413**	0.2595
TRV	0.1615	0.1726	0.1762	0.1513	0.1495	**0.1431**	0.2865
TSLA	0.1585	**0.1437**	0.2219	0.1931	0.1950	0.1746	0.2378
UNH	0.1972	0.1961	0.2286	0.2349	0.2334	**0.1785**	0.2883
V	1.1644	0.2749	0.2398	0.2532	0.2574	**0.2197**	0.3390
VZ	1.5620	0.1697	0.1933	0.1731	0.1814	**0.1525**	0.2606
WMT	0.9095	0.2674	0.2456	0.2355	0.2353	**0.2188**	0.2985
XOM	2.4919	0.3536	0.2074	0.1605	**0.1594**	0.1612	0.3179
Average	2.7012	0.2312	0.2438	0.2133	0.2159	**0.1883**	0.3048
FX							
AUDUSD	0.1179	0.1173	0.1485	**0.1012**	0.1039	0.2264	0.1806
EURUSD	0.1039	**0.1003**	0.1416	0.1005	0.1024	0.2346	0.1678
GBPUSD	0.1762	**0.0998**	0.1655	0.1167	0.1176	0.2573	0.1877
USDCAD	0.0914	0.0879	0.1142	**0.0828**	0.0844	0.1897	0.1487
USDJPY	0.2233	**0.1953**	0.3177	0.2375	0.2388	0.3421	0.2867
Average	0.1425	**0.1201**	0.1775	0.1277	0.1294	0.2500	0.1943
Futures							
C	0.1985	**0.1772**	0.2020	0.2159	0.2212	0.1851	0.2632
CL	0.2014	**0.1645**	0.2832	0.1672	0.1746	0.1772	0.4519
ES	1.4887	**0.2531**	0.4722	0.2805	0.2751	0.3454	0.5195
GC	0.1216	**0.1057**	0.2129	0.1297	0.1727	0.1952	0.1886
NG	0.1599	**0.1231**	0.1729	0.1322	0.1354	0.1301	0.2409
Average	0.4340	**0.1647**	0.2687	0.1851	0.1958	0.2066	0.3328
Overall Average	2.2186	0.2134	0.2397	0.2019	0.2052	**0.1963**	0.2965

Per-asset QLIKE loss at $h = 5$. QLIKE is computed as $\text{mean}(RV_t/\hat{RV}_t - \ln(RV_t/\hat{RV}_t) - 1)$. Bold indicates the lowest (best) QLIKE in each row. Chr-Bolt-S = Chronos-Bolt-Small, Chr-Bolt-B = Chronos-Bolt-Base, Moirai-S = Moirai-2.0-Small. HAR-J, HAR-RS, and HARQ are omitted as their QLIKE values are dominated by near-zero forecast artifacts (see Section 5.1).

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	Lag-Llama
Stocks							
AAPL	2.0348	0.2594	0.2747	0.3120	0.3171	**0.2428**	0.2482
ADBE	3.5556	0.2147	0.2431	0.2394	0.2449	0.2022	**0.1963**
AMD	0.1516	0.1341	0.1681	0.1313	0.1344	0.1307	**0.1185**
AMGN	3.2254	0.2634	**0.2177**	0.3563	0.3541	0.2988	0.2610
AMZN	23.0166	0.4877	0.4683	0.5492	0.5540	0.4759	**0.4003**
AXP	4.1452	0.3781	0.3191	0.3952	0.3945	**0.2702**	0.3868
BA	48.4755	1.1024	0.4859	0.4640	0.4634	**0.4147**	0.4528
CAT	4.6059	0.2131	0.1892	0.2444	0.2438	**0.1883**	0.1931
CRM	11.4276	0.3410	0.2683	0.2891	0.2881	0.2422	**0.2096**
CSCO	2.6118	0.2628	0.3505	0.2953	0.2982	**0.2257**	0.2550
CVX	16.7375	0.7459	0.3206	0.3418	0.3423	**0.2794**	0.3964
DIS	16.3347	0.5921	**0.1851**	0.3167	0.3324	0.2301	0.3033
GE	**0.1934**	0.2004	0.2111	0.2373	0.2380	0.1991	0.2203
GOOGL	6.7952	0.2986	**0.2545**	0.3646	0.3818	0.3120	0.2549
GS	0.2048	0.2145	0.2868	0.2503	0.2558	**0.2007**	0.2303
HD	1.4765	0.3321	0.3579	0.3579	0.3567	**0.2690**	0.3500
HON	6.7514	0.5217	0.3450	0.3785	0.3761	**0.2563**	0.3582
IBM	2.1795	0.2645	0.2330	0.3178	0.3269	**0.2153**	0.2770
JNJ	0.3284	**0.2763**	0.2872	0.3486	0.3507	0.2890	0.2817
JPM	6.3848	0.3178	0.3438	0.3719	0.3691	**0.2749**	0.3335
KO	3.2457	0.3462	0.3343	0.3918	0.3967	**0.2948**	0.3511
MCD	4.4248	0.4672	0.3247	0.4223	0.4230	**0.3013**	0.4126
META	4.1675	0.2438	0.2527	0.2389	0.2411	0.2123	**0.2056**
MMM	1.9908	0.2210	**0.2090**	0.2714	0.2690	0.2264	0.2122
MRK	0.2130	0.1894	**0.1718**	0.2568	0.2582	0.1788	0.2079
MSFT	4.6384	**0.2857**	0.3099	0.3927	0.4017	0.3111	0.3071
NFLX	9.5100	0.3088	**0.2297**	0.3269	0.3278	0.3115	0.2546
NKE	2.9300	0.3044	0.2201	0.2841	0.2867	**0.2167**	0.2653
NVDA	0.2246	0.2052	0.2805	0.2322	0.2359	0.2086	**0.2019**
ORCL	7.1831	0.3691	0.3904	0.4655	0.4867	0.3851	**0.3491**
PG	3.1120	0.3395	0.3016	0.3307	0.3344	**0.2721**	0.3286
PM	0.2639	**0.2057**	0.2625	0.2900	0.2941	0.2525	0.2789
SHW	5.1294	0.2928	**0.1803**	0.2545	0.2588	0.2181	0.2464
TRV	0.2825	0.3321	0.2575	0.2930	0.2850	**0.2293**	0.2953
TSLA	**0.1561**	0.1660	0.1848	0.2469	0.2520	0.2213	0.1921
UNH	3.1724	0.2263	**0.2006**	0.3292	0.3209	0.2221	0.2567
V	3.8648	0.4143	**0.2704**	0.3721	0.3807	0.2710	0.3090
VZ	1.9827	0.2200	0.2335	0.2702	0.2756	**0.2068**	0.2390
WMT	6.0957	0.4762	0.2769	0.3359	0.3330	**0.2550**	0.2697
XOM	19.0392	0.9555	0.3003	0.3194	0.3247	**0.2565**	0.3941
Average	6.0566	0.3497	0.2750	0.3222	0.3252	**0.2567**	0.2826
FX							
AUDUSD	0.1665	0.1624	0.1645	**0.1441**	0.1478	0.2003	0.1739
EURUSD	0.1267	0.1306	0.1469	**0.1232**	0.1284	0.2007	0.1409
GBPUSD	0.2949	**0.1254**	0.1719	0.1479	0.1485	0.2240	0.1533
USDCAD	0.1115	**0.1092**	0.1240	0.1158	0.1196	0.1637	0.1329
USDJPY	0.2295	**0.2081**	0.3196	0.2741	0.2714	0.2701	0.2166
Average	0.1858	**0.1471**	0.1854	0.1610	0.1631	0.2117	0.1635
Futures							
C	0.1695	**0.1435**	0.1502	0.2023	0.2040	0.1563	0.1853
CL	0.3207	0.4053	0.4021	0.2964	0.2970	**0.2697**	0.5906
ES	4.5906	0.6009	0.5021	0.5946	0.6009	**0.4383**	0.5478
GC	0.1340	**0.1209**	0.2085	0.1683	0.1770	0.1760	0.1553
NG	0.1661	**0.1317**	0.1653	0.1477	0.1520	0.1478	0.2034
Average	1.0762	0.2805	0.2857	0.2819	0.2862	**0.2376**	0.3365
Overall Average	4.9714	0.3226	0.2671	0.3020	0.3051	**0.2503**	0.2761

Per-asset QLIKE loss at $h = 22$. QLIKE is computed as $\text{mean}(RV_t/\hat{RV}_t - \ln(RV_t/\hat{RV}_t) - 1)$. Bold indicates the lowest (best) QLIKE in each row. Chr-Bolt-S = Chronos-Bolt-Small, Chr-Bolt-B = Chronos-Bolt-Base, Moirai-S = Moirai-2.0-Small. HAR-J, HAR-RS, and HARQ are omitted as their QLIKE values are dominated by near-zero forecast artifacts (see Section 5.1).

Model	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	p(joint)	% Reject
Econometric benchmarks						
HAR	0.0000	1.155	0.276	7.09	0.261	28%
HAR-J	0.0000	1.154	0.293	5.99	0.268	28%
HAR-RS	0.0001	0.910	0.210	15.64	0.143	52%
HARQ	0.0001	0.681	0.119	13.01	0.063	70%
Log-HAR	−0.0000	1.438	0.198	5.42	0.050	76%
ARFIMA	0.0001	0.977	0.221	15.90	0.099	72%
Foundation models						
Chronos-Bolt-S	0.0000	1.282	0.400	26.29	0.002	98%
Chronos-Bolt-B	0.0000	1.438	0.417	31.24	0.001	98%
Moirai-2.0-S	0.0000	1.094	0.367	18.12	0.007	94%
Lag-Llama	0.0000	1.183	0.069	5.01	0.055	60%

Table 20: Mincer–Zarnowitz regression ($h = 5$)

Mincer–Zarnowitz forecast efficiency regression, $h = 5$. See Table 12 for variable definitions.

Model	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	p(joint)	% Reject
Econometric benchmarks						
HAR	0.0001	0.490	0.096	18.82	0.044	70%
HAR-J	0.0001	0.510	0.102	16.83	0.044	70%
HAR-RS	0.0002	0.366	0.083	32.16	0.016	92%
HARQ	0.0002	0.299	0.074	54.19	0.003	100%
Log-HAR	0.0001	0.474	0.072	16.70	0.018	92%
ARFIMA	0.0001	0.578	0.143	16.87	0.040	80%
Foundation models						
Chronos-Bolt-S	0.0001	1.121	0.202	10.98	0.008	96%
Chronos-Bolt-B	0.0000	1.453	0.221	12.15	0.005	98%
Moirai-2.0-S	0.0000	0.985	0.187	12.55	0.006	98%
Lag-Llama	0.0001	0.705	0.057	3.43	0.096	50%

Table 21: Mincer–Zarnowitz regression ($h = 22$)

Mincer–Zarnowitz forecast efficiency regression, $h = 22$. See Table 12 for variable definitions.

	MSE ($\times 10^8$)		QLIKE		R^2_{OOS}	
Model	252	512	252	512	252	512
HAR	14.95	**12.32**	2.4626	**0.1794**	0.221	**0.389**
HAR-J	14.65	**11.90**	2.2567	**0.2355**	0.242	**0.407**
HAR-RS	17.09	**12.84**	6.3478	**0.4547**	0.108	**0.360**
HARQ	17.54	**14.45**	14.4245	**4.7712**	0.040	**0.240**
Log-HAR	14.94	**13.38**	0.2211	**0.1628**	0.156	**0.316**
ARFIMA	**16.37**	29.68	**0.2551**	0.4191	**0.144**	-0.659

Table 22: Estimation window sensitivity ($h = 5$)

Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 5$, 40 equities). Cross-asset mean metrics. Bold indicates the better window for each model.

	MSE ($\times 10^8$)		QLIKE		R^2_{OOS}	
Model	252	512	252	512	252	512
HAR	10.71	**9.33**	5.3791	**0.2744**	-0.011	**0.145**
HAR-J	10.66	**9.25**	5.2619	**0.3315**	0.004	**0.154**
HAR-RS	11.88	**9.98**	8.4756	**0.7274**	-0.126	**0.089**
HARQ	11.95	**10.05**	10.9013	**3.3530**	-0.158	**0.066**
Log-HAR	10.42	**9.29**	0.3247	**0.1966**	-0.017	**0.149**
ARFIMA	**11.47**	26.51	**0.2757**	0.4123	**-0.058**	-1.564

Table 23: Estimation window sensitivity ($h = 22$)

Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 22$, 40 equities). Cross-asset mean metrics. Bold indicates the better window for each model.